

Kratos Multiphysics for Computational Fluid Dynamics

Kratos Workshop 2024 - Padova

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Overview

- Incompressible flow
 - Main features
 - Unfitted mesh methods
- Potential flow
- Compressible flow
- Shallow water approximation
- 2-phase and free-surface flow
- URANS and optimisation
- Other technologies
- Coupling
 - Fluid-Structure Interaction
 - Thermal
- Ongoing research & developments

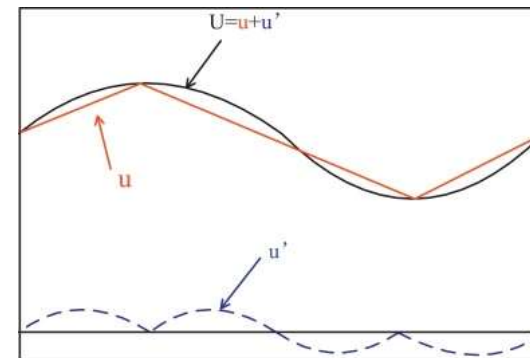
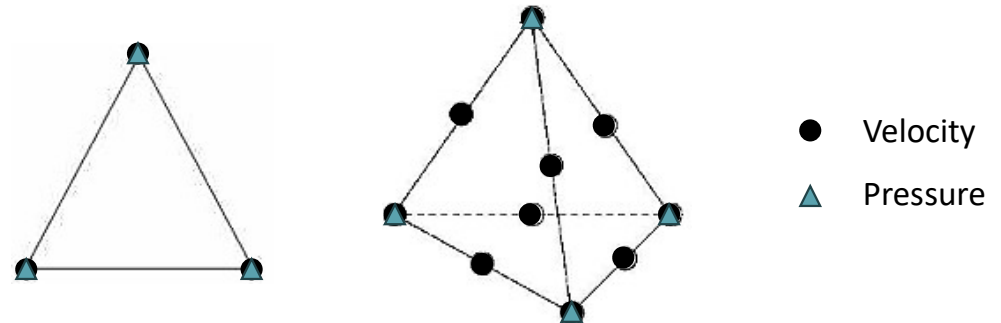


Incompressible flow



Incompressible flow: main features

- Available interpolations:
 - Equal-order: linear and bilinear elements
 - Div-stable: P2P1 element (Taylor-Hood)
- Stabilisation techniques
 - **Variational MultiScales paradigm (VMS)**
 - Algebraic SubGrid Scales (ASGS)
 - Orthogonal SubGrid Scales (OSGS)
 - Other: Finite Increment Calculus (FIC)
- **Turbulence** modelling
 - VMS stabilisation → understood as LES
 - Smagorinsky model
 - Spallart-Allmaras model
- **Newtonian** and **non-Newtonian** constitutive models
- Support for **mesh motion** (ALE) → `MeshMovingApplication`



VMS paradigm (from Z.-J. Chen et al. 2018)

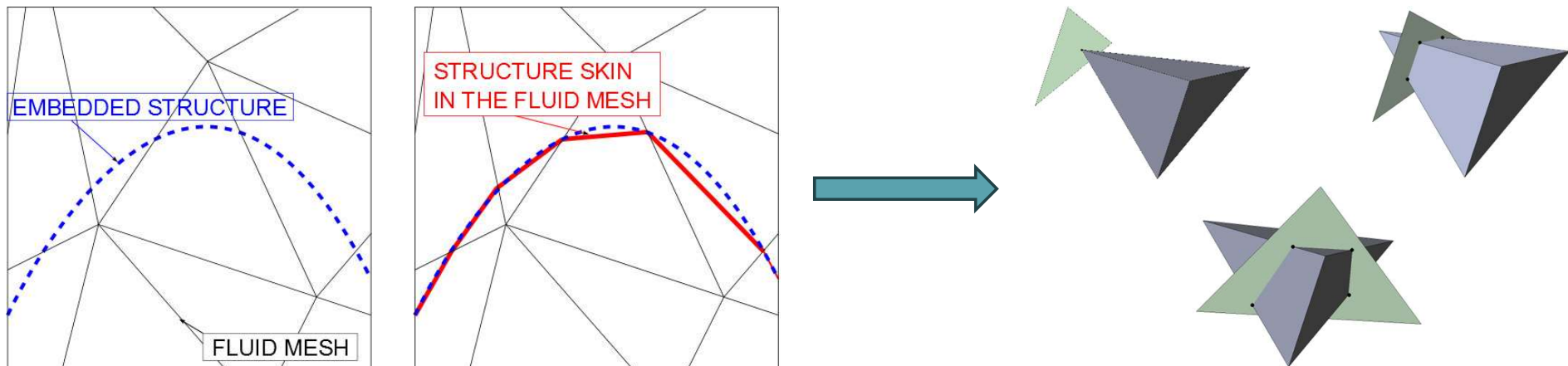


Incompressible flow: unfitted mesh methods

Main idea: decouple mesh boundaries from computational ones

- **Background mesh:** actual computational mesh
- **Skin mesh:** auxiliary mesh representing the boundaries of the immersed object

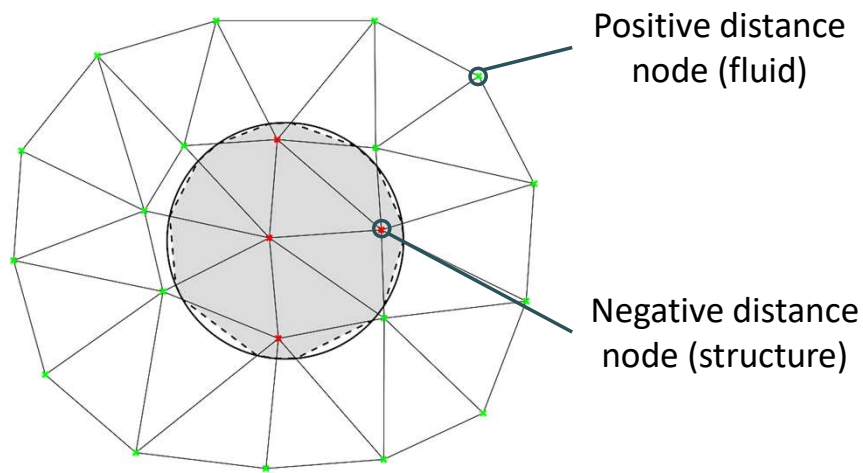
Carve out “holes” in the background mesh → represent the immersed object from the **intersections**



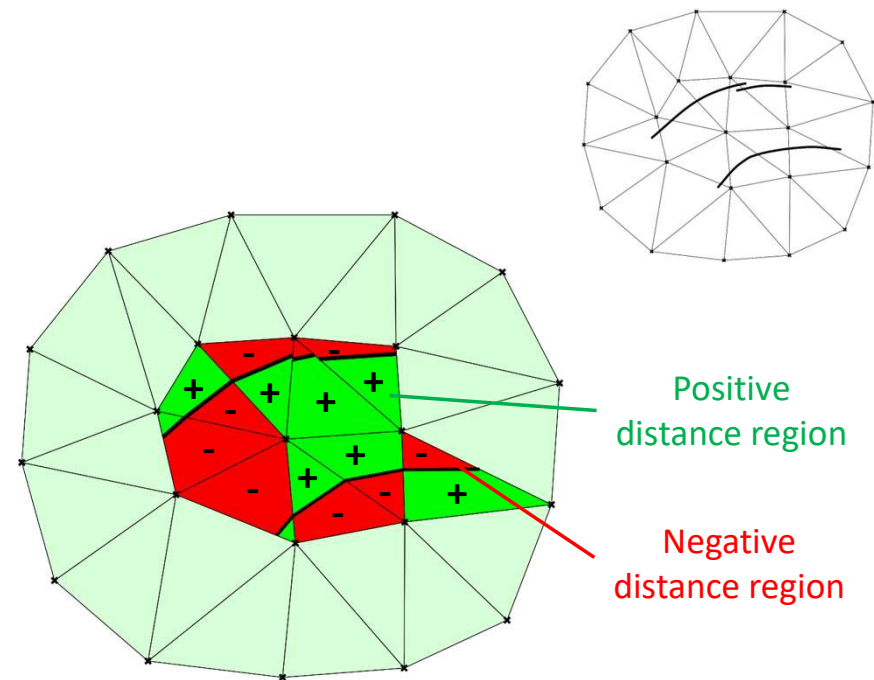
Incompressible flow: unfitted mesh methods

Represent the immersed object from the **intersections**:

- **Level set** description → CutFEM



Continuous distance (volumetric bodies)



Discontinuous distance (thin-walled bodies)

- Directly use the **intersections** → SBM (ongoing)



Incompressible flow: unfitted mesh methods

Why unfitted mesh discretisations?

1. Cope with **arbitrary boundary movements**

- Avoid element distortion (or inversion)
- Topology changes → avoid remeshing

2. Handle **“dirty” input geometries** (gaps, overlaps, ...)

- *“clean up operations can reach the 50% of the total simulation time”* (Cottrell et al., 2009)

3. Efficient **thin-walled bodies pre-processing**:

- Avoid creating a volume mesh from a thin-walled body (manual surface duplication, error prone for meshing alg.)



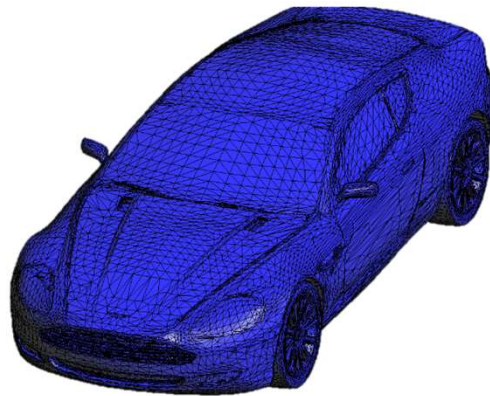
Incompressible flow: unfitted mesh methods

1. Cope with **arbitrary boundary movements**: moving cylinder in channel



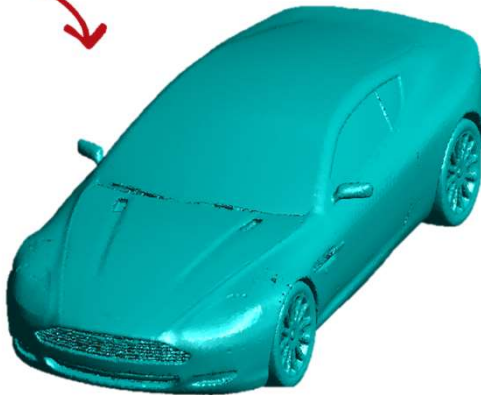
Incompressible flow: unfitted mesh methods

2. Handle “dirty” input geometries (gaps, overlaps, ...): Aston Martin Vanquish .stl input file

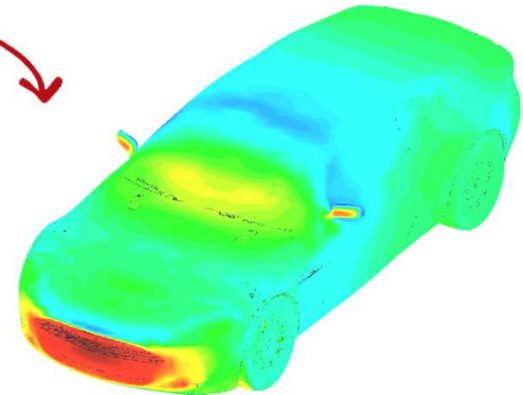


Input .stl

No clean
up needed!



Level set representation



Pressure field



Incompressible flow: unfitted mesh methods

2. Handle “dirty” input geometries (gaps, overlaps, ...): Montigalà district wind simulation

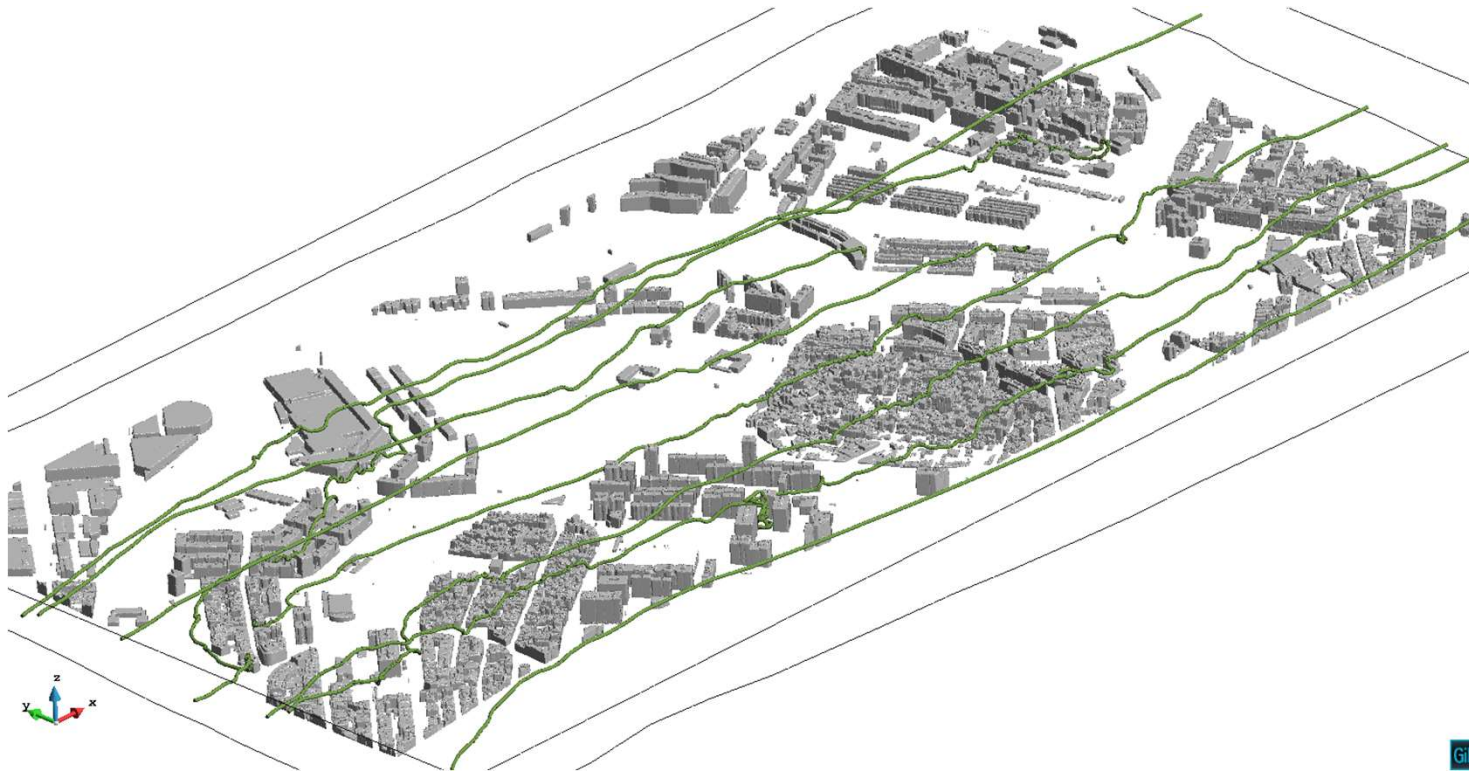
- Geometry:
 - Ground: body-fitted from cartography
 - Buildings: level set representation from public data (1m resolution)
- 55M tetrahedra unstructured mesh
- 192 MPI cores

Domain of interest



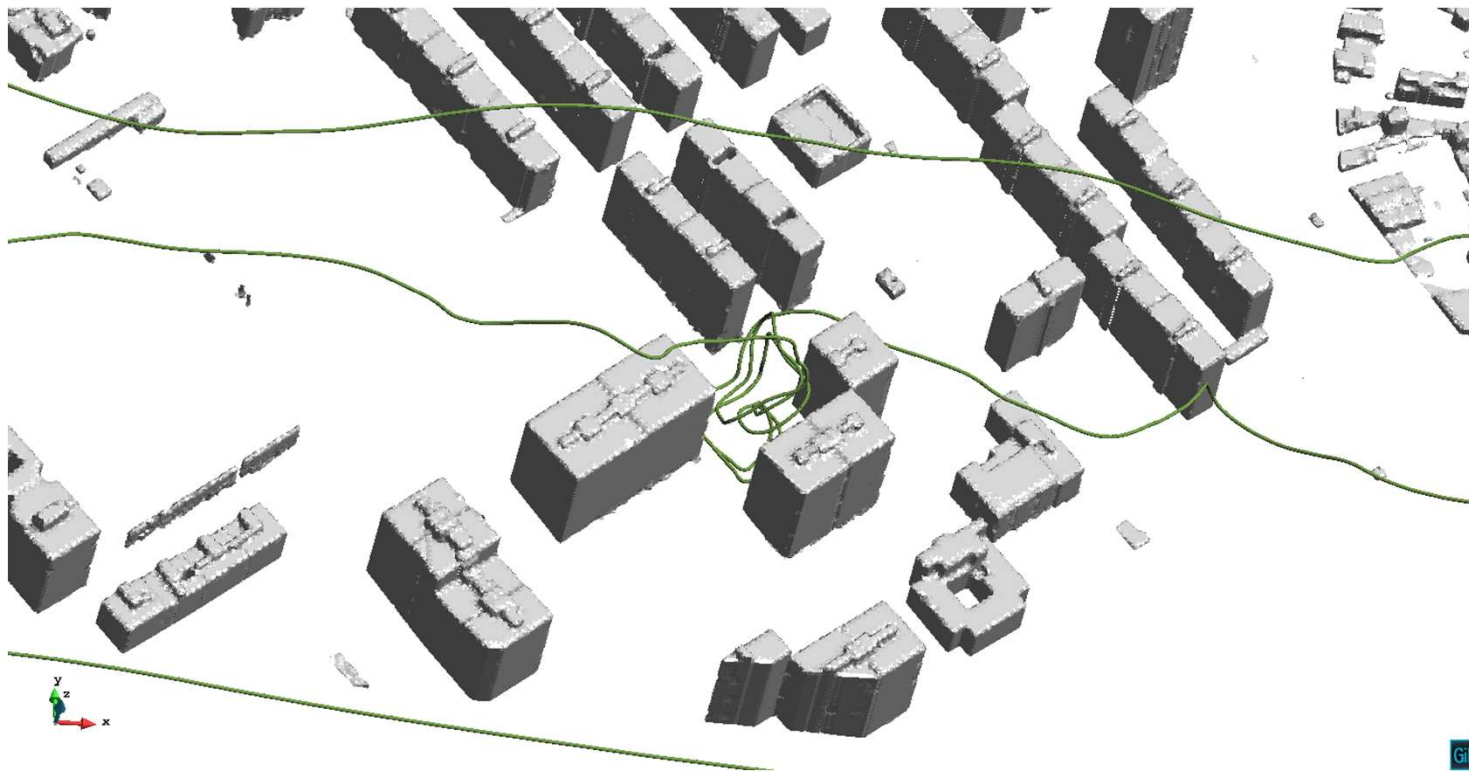
Incompressible flow: unfitted mesh methods

2. Handle “dirty” input geometries (gaps, overlaps, ...): Montigalà district wind simulation



Incompressible flow: unfitted mesh methods

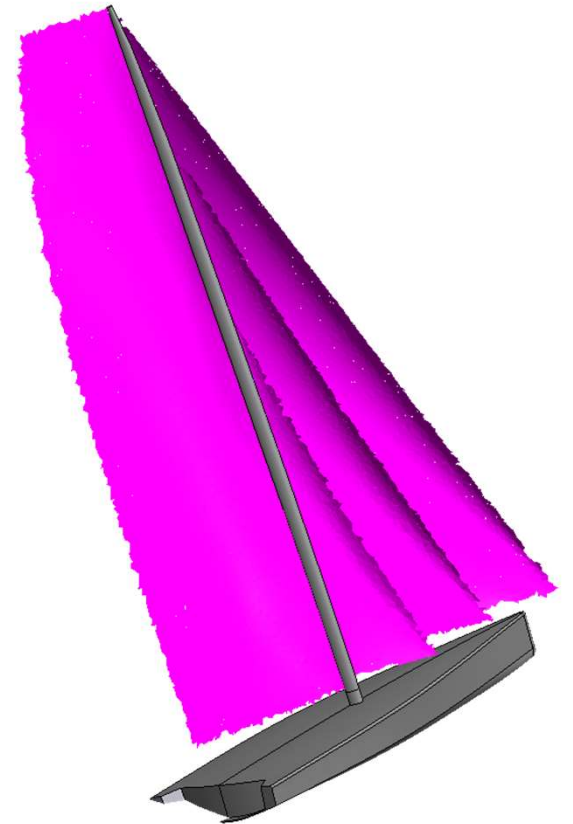
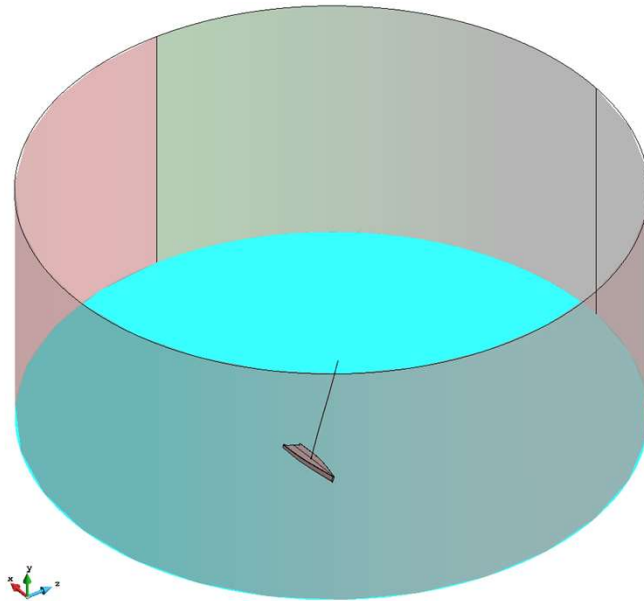
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Incompressible flow: unfitted mesh methods

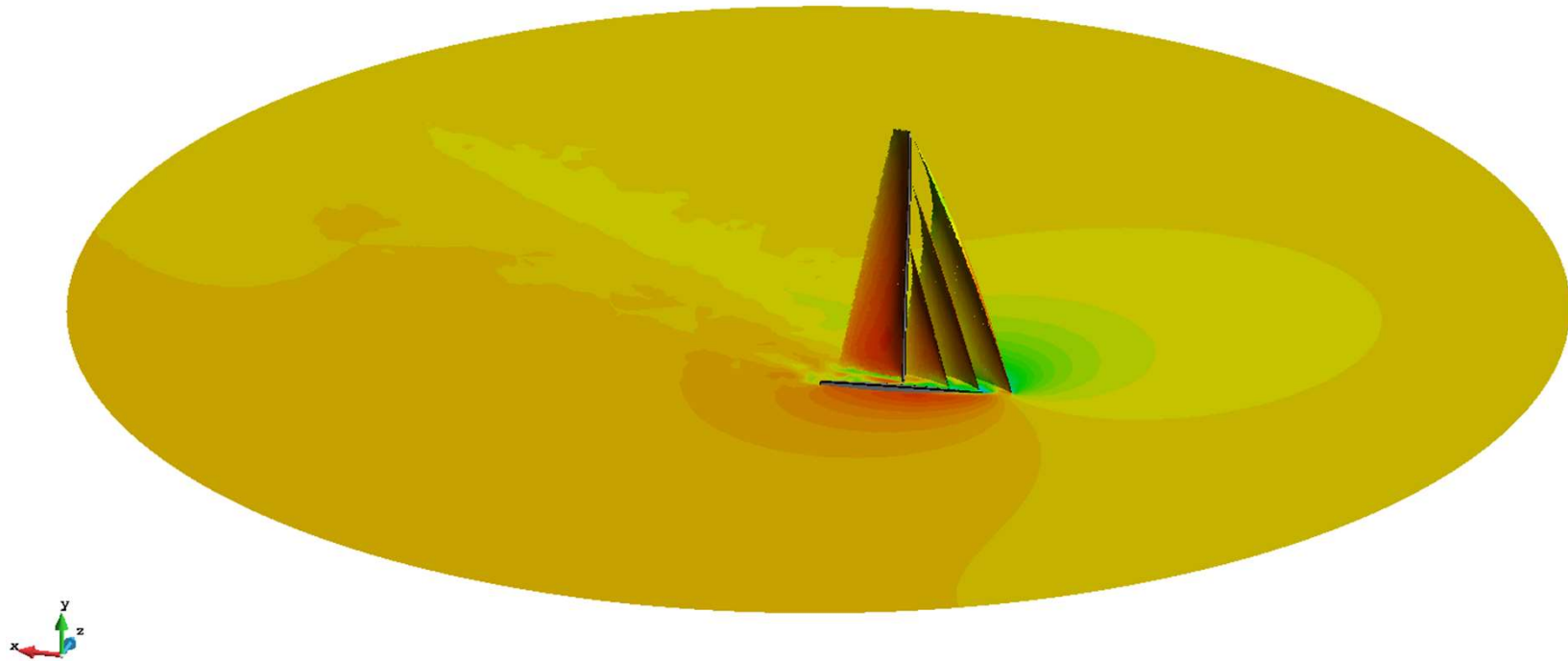
3. Efficient **thin-walled bodies pre-processing**: flow around a sailboat

- 80m radius and 70m height cylinder computational domain
- 3,3M tetrahedra unstructured mesh
- 15knots (7,7m/s) inlet (rotated 45°)
- **Hull and mast → Body fitted**
- **Sails → Embedded**



Incompressible flow: unfitted mesh methods

3. Efficient **thin-walled bodies pre-processing**: flow around a sailboat

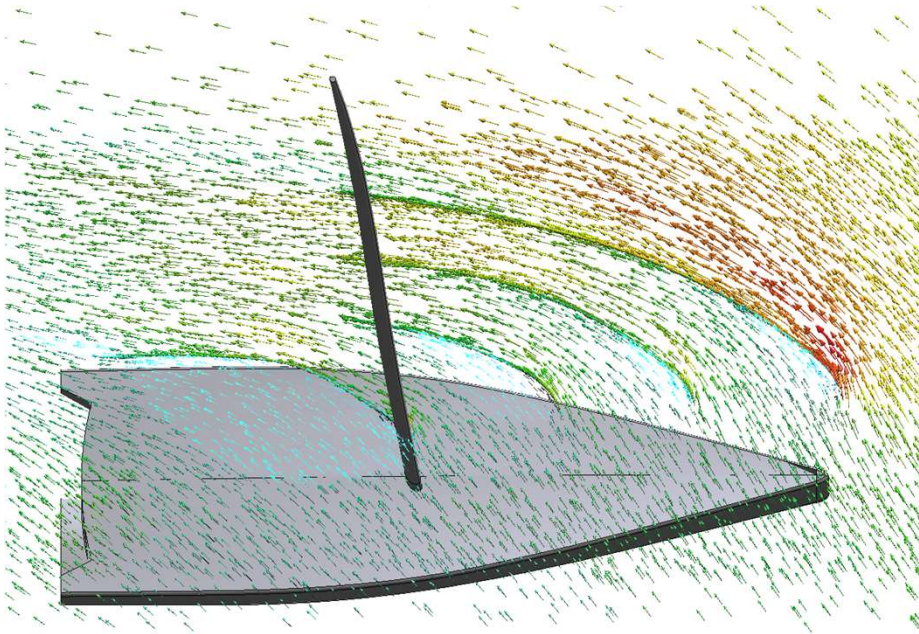


Pressure field cross section

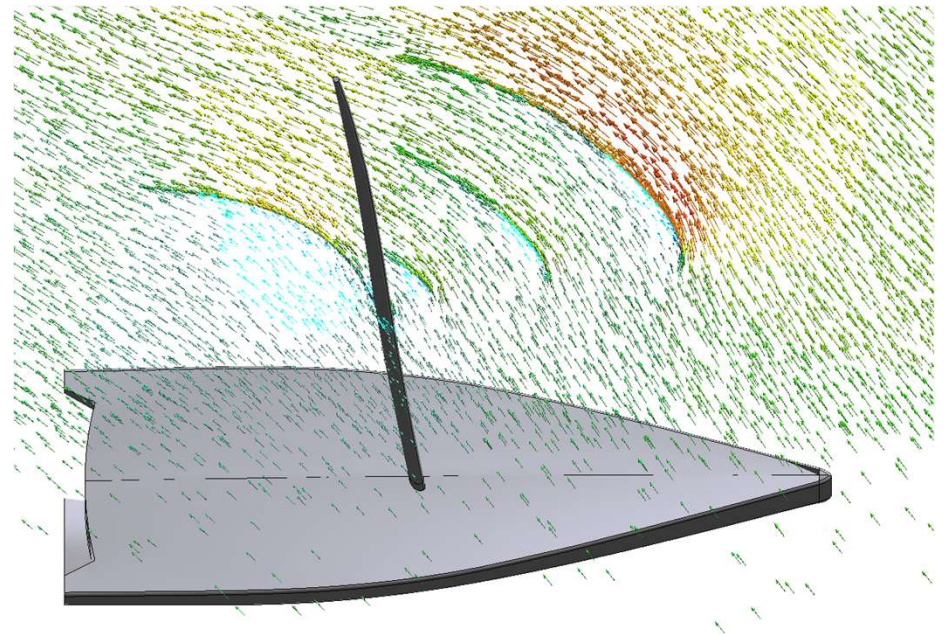


Incompressible flow: unfitted mesh methods

3. Efficient **thin-walled bodies pre-processing**: flow around a sailboat



5m cross section (velocity vector field)

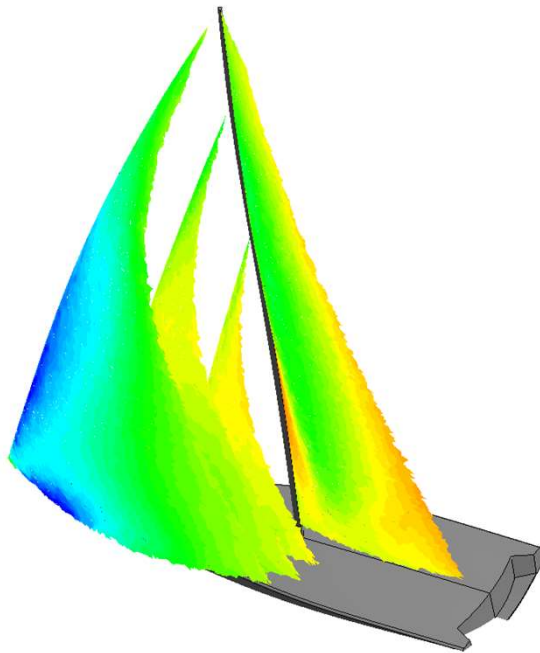


12.5m cross section (velocity vector field)

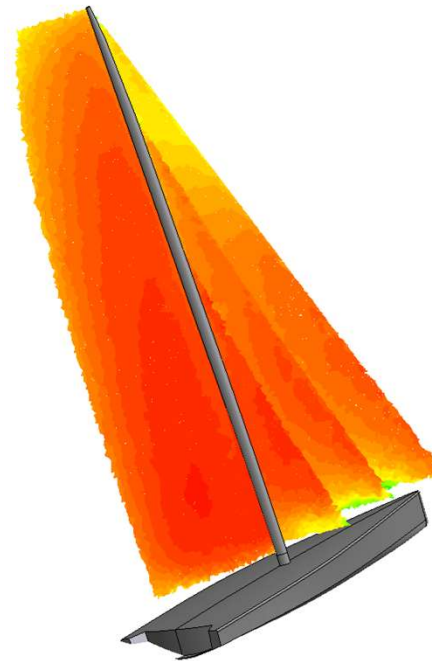


Incompressible flow: unfitted mesh methods

3. Efficient **thin-walled bodies pre-processing**: flow around a sailboat



Pressure field (leeward side)



Pressure field (windward side)

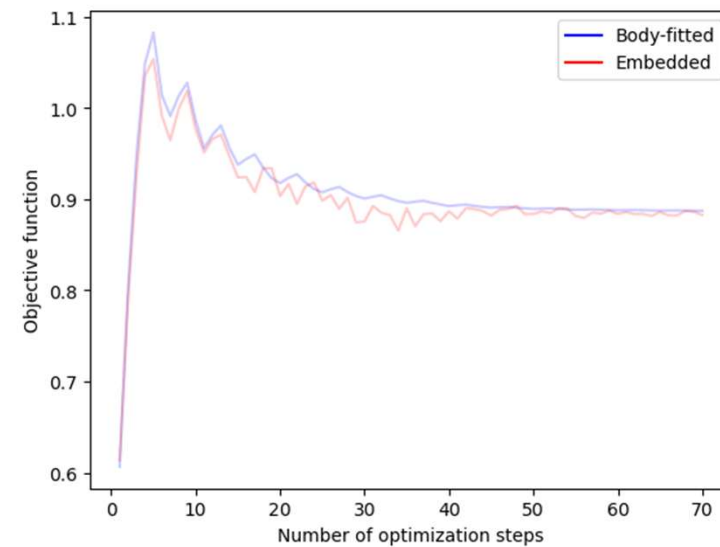
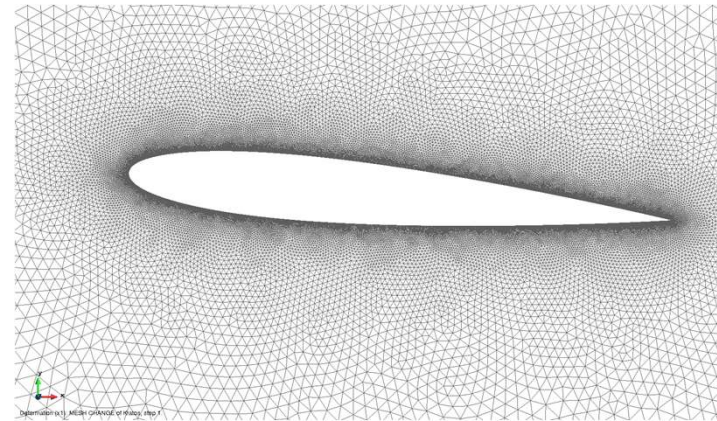


Potential flow



Potential flow

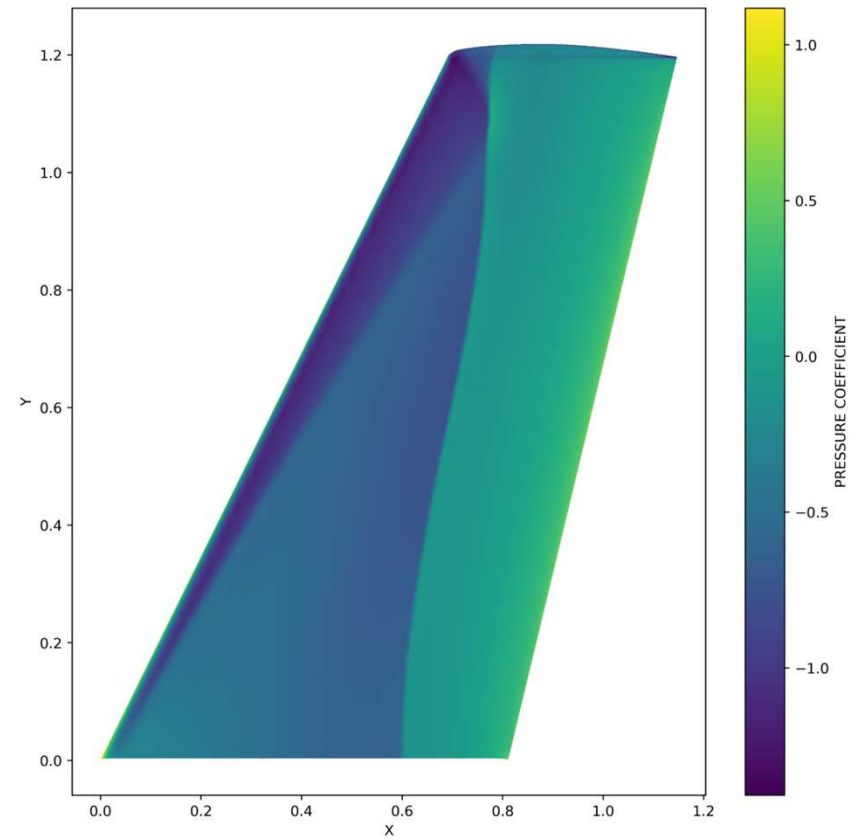
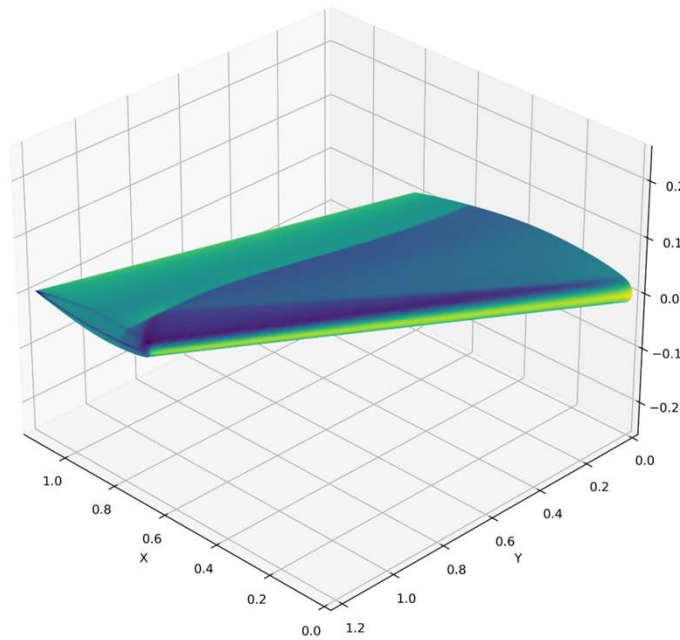
- Available formulations:
 - Incompressible
 - Compressible
 - Transonic (compressible + artificial viscosity)
- **Body-fitted** and **unfitted** discretisations
- **Unfitted wake** for Kutta condition
- **Shape optimization**
 - Adjoints calculation
 - Unfitted discretization



Potential flow

Onera M6 wing validation

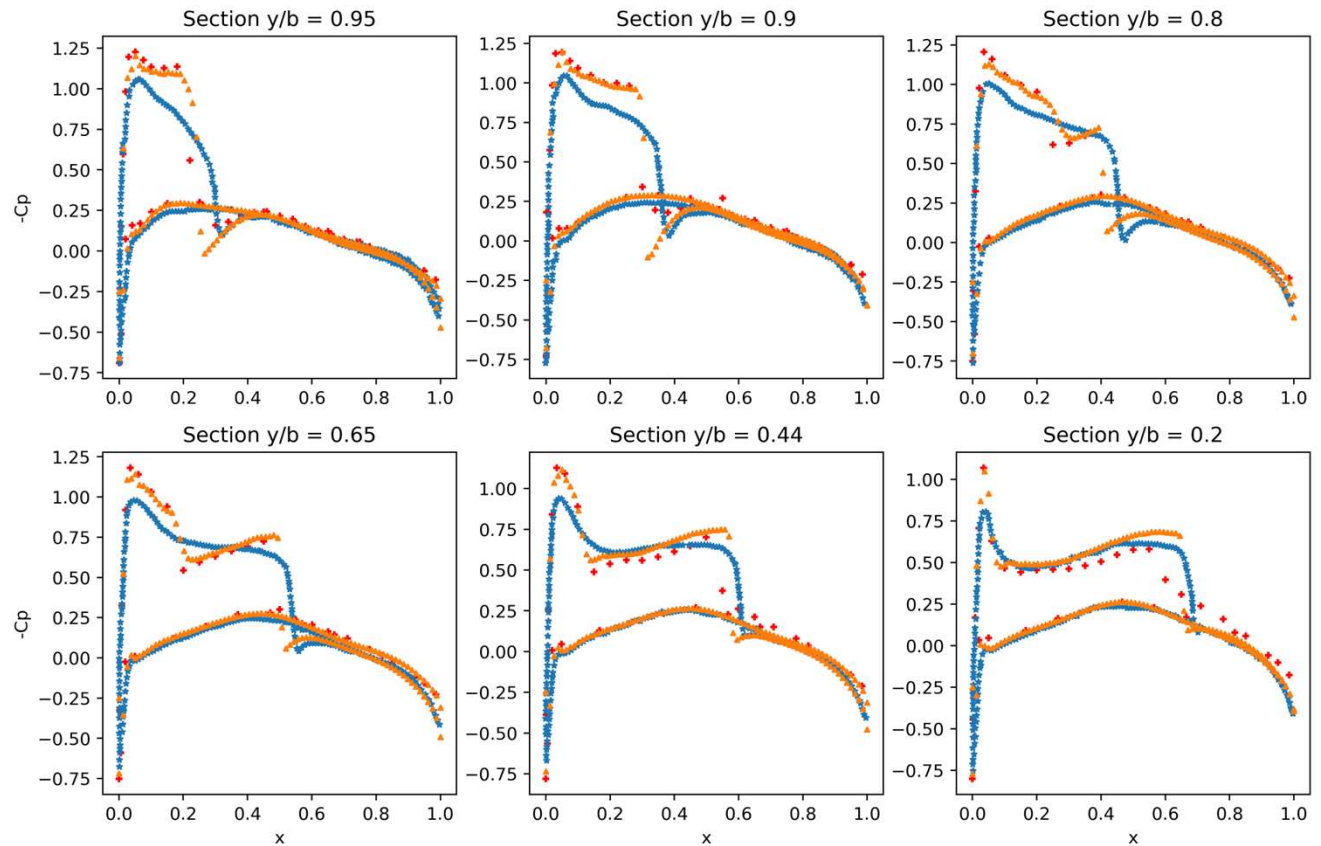
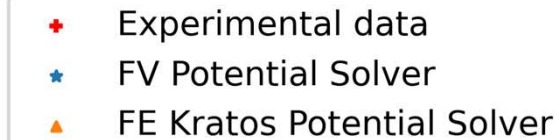
- Angle of attack: $3,06^\circ$
- Mach: 0,839



Potential flow

Onera M6 wing validation

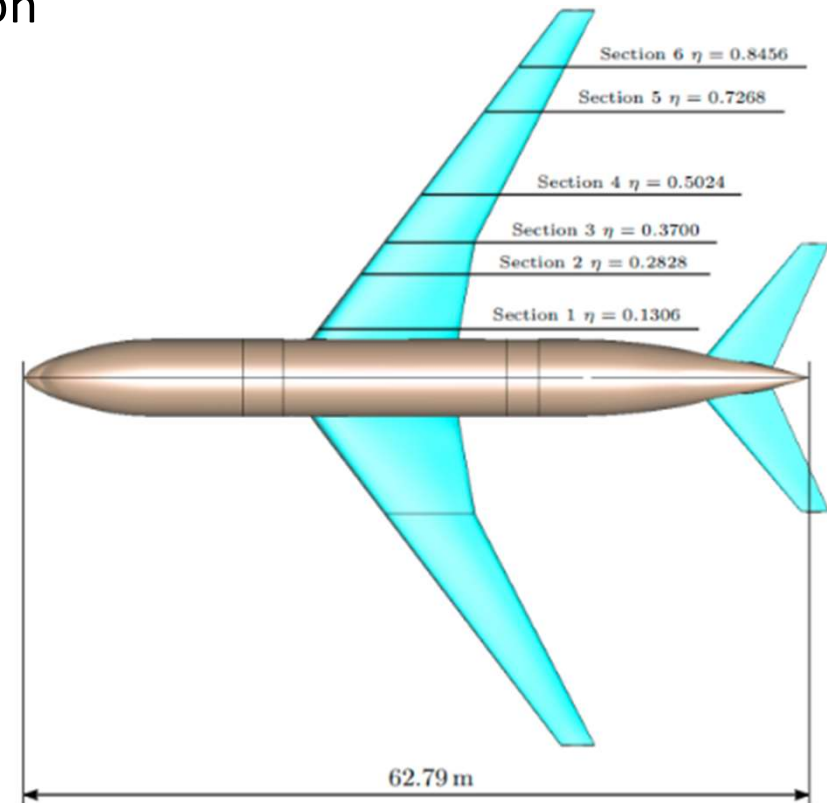
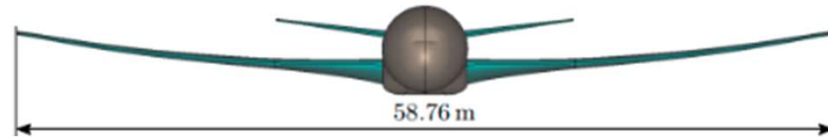
- Angle of attack: $3,06^\circ$
- Mach: 0,839



Potential flow

Boeing and NASA CRM wing-body-tail validation

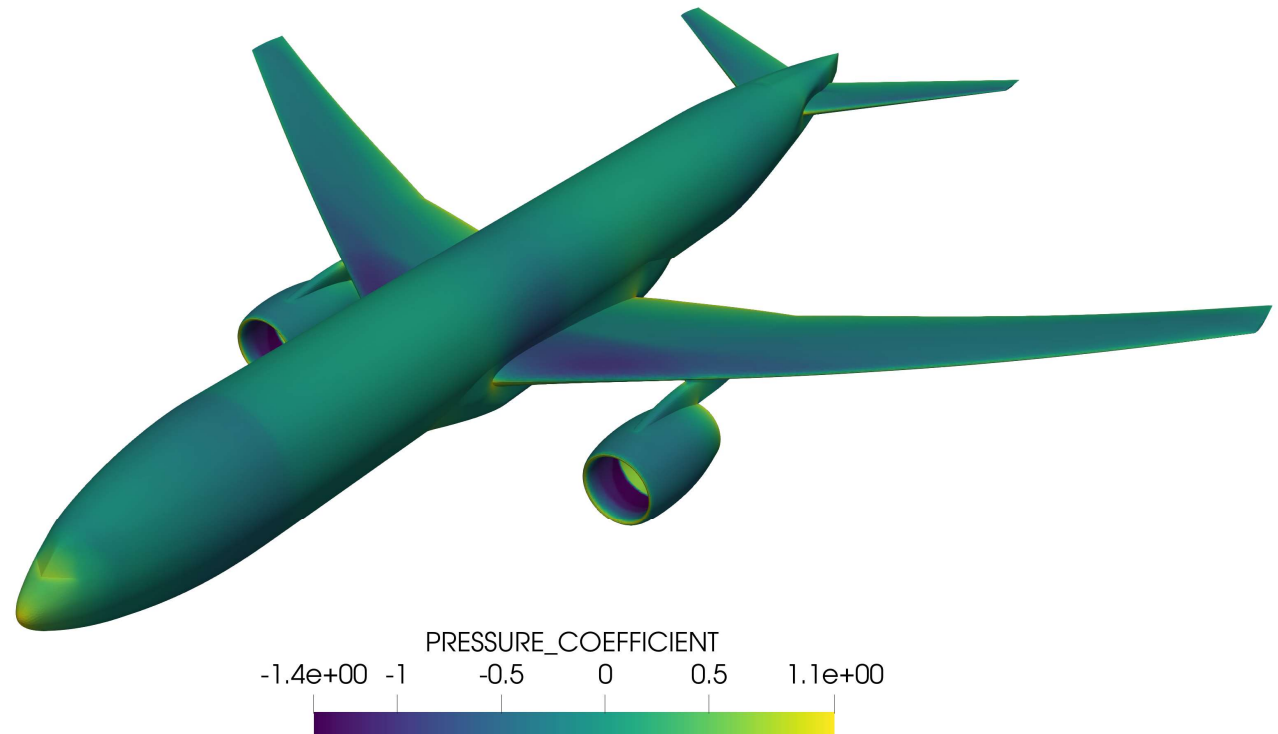
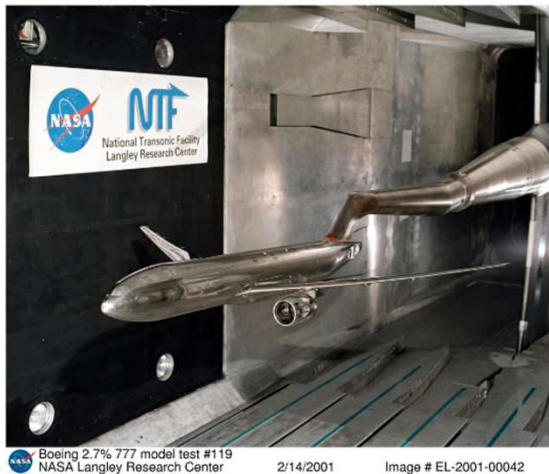
- Angle of attack: $2,31^\circ$
- Mach: 0,85



Potential flow

Boeing and NASA CRM wing-body-tail validation

- Angle of attack: $2,31^\circ$
- Mach: 0,85



Compressible flow



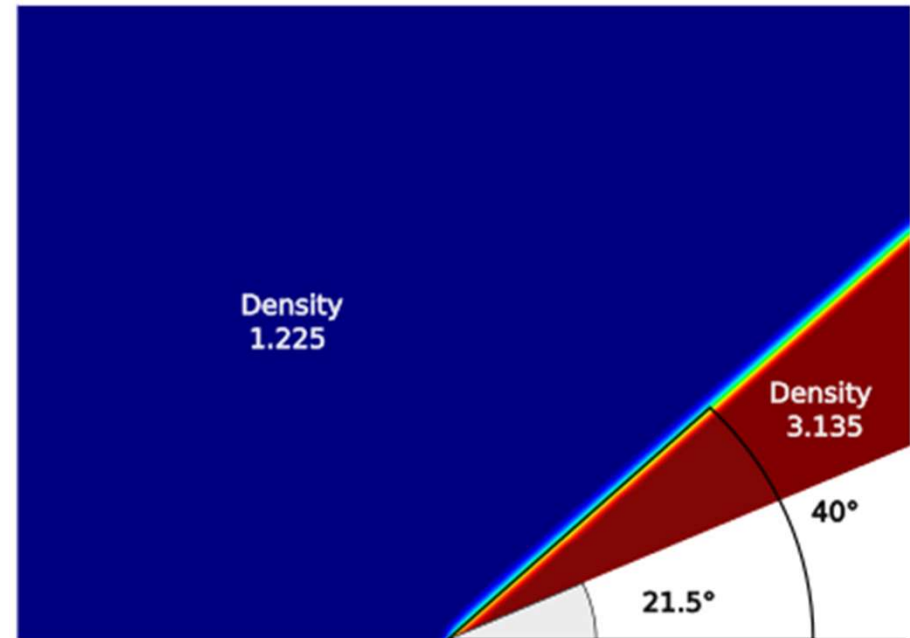
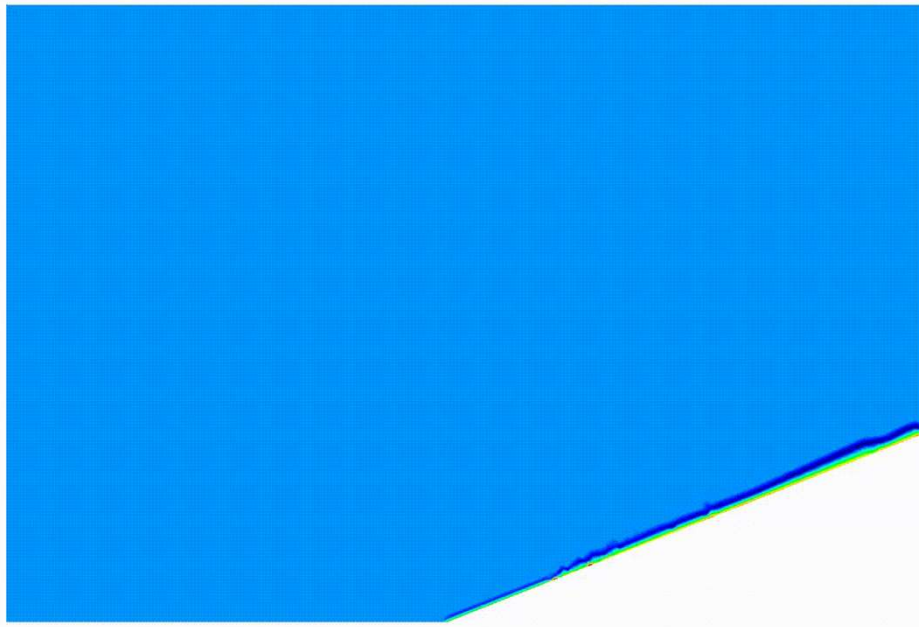
Compressible flow

- **Energy coupled formulation** written in conservative variables
 - Momentum
 - Density
 - Total energy
- **VMS stabilization** (ASGS and OSS)
- **Explicit time integration**
 - RK-type schemes (RK3-TVD, RK4)
- **Shock capturing** techniques
 - Physics-based (Peraire et al., 2018)
 - Entropy-based (Guermond et al., 2011)



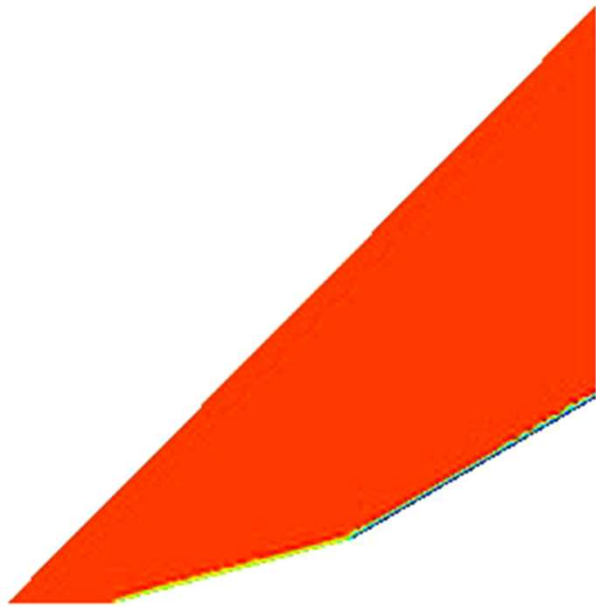
Compressible flow

Supersonic flow over a wedge

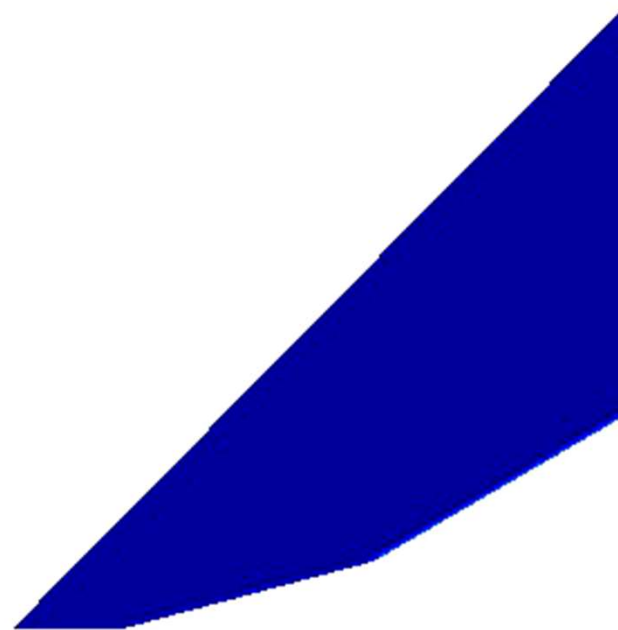


Compressible flow

Supersonic flow over a double wedge



Mach number

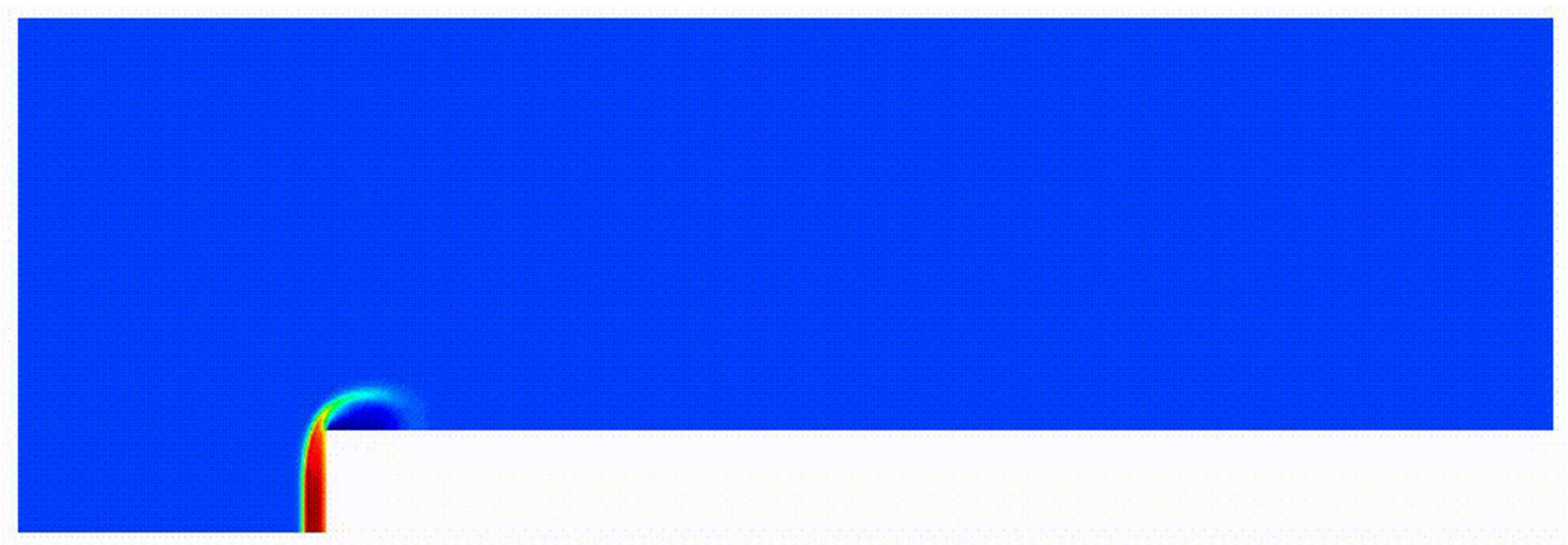


Pressure



Compressible flow

Supersonic flow past a step

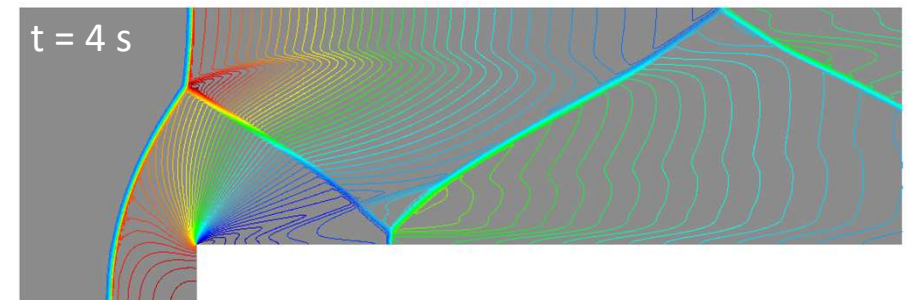
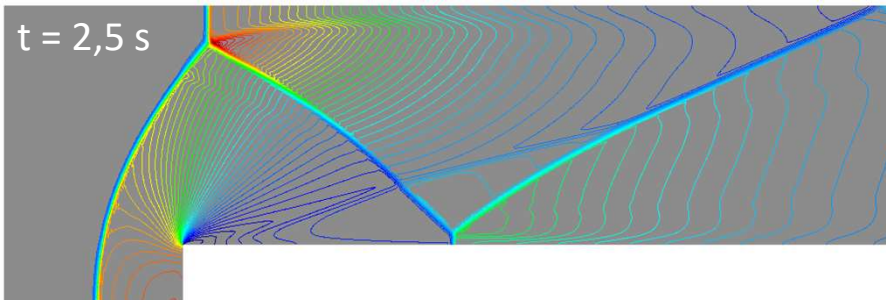
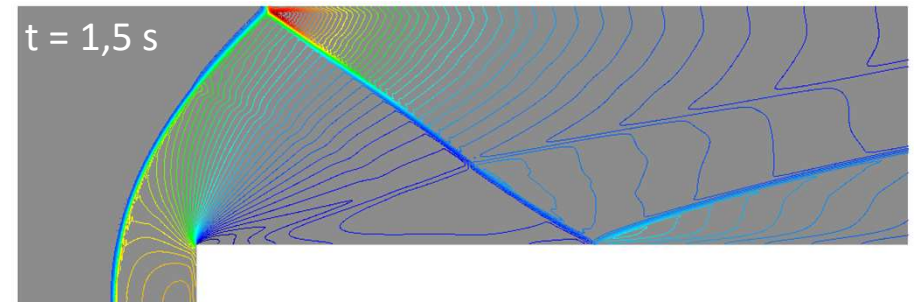
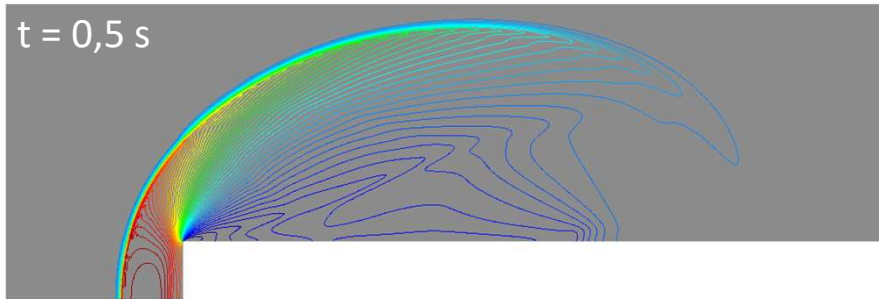


Density contour field



Compressible flow

Supersonic flow past a step



Density contour lines

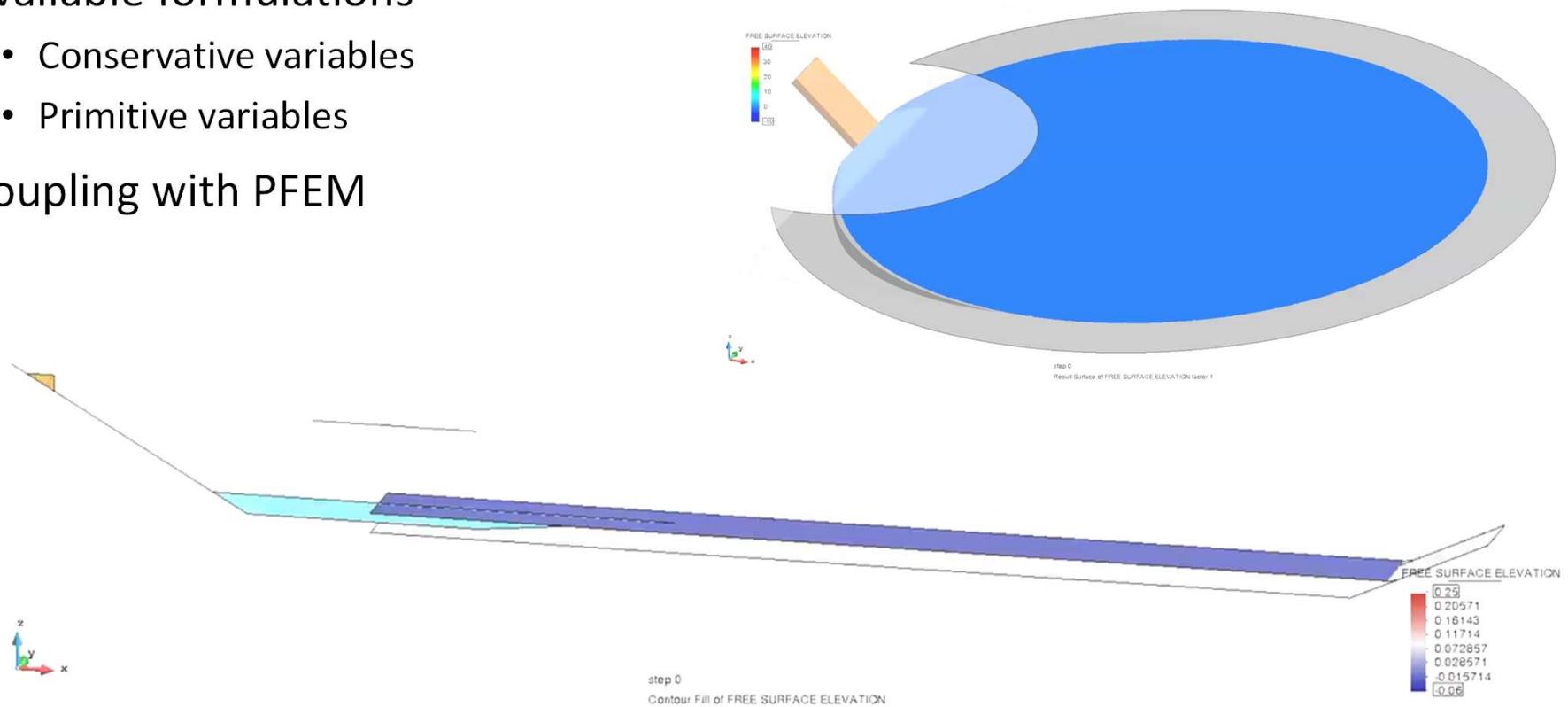


Shallow water approximation



Shallow water approximation

- Available formulations
 - Conservative variables
 - Primitive variables
- Coupling with PFEM



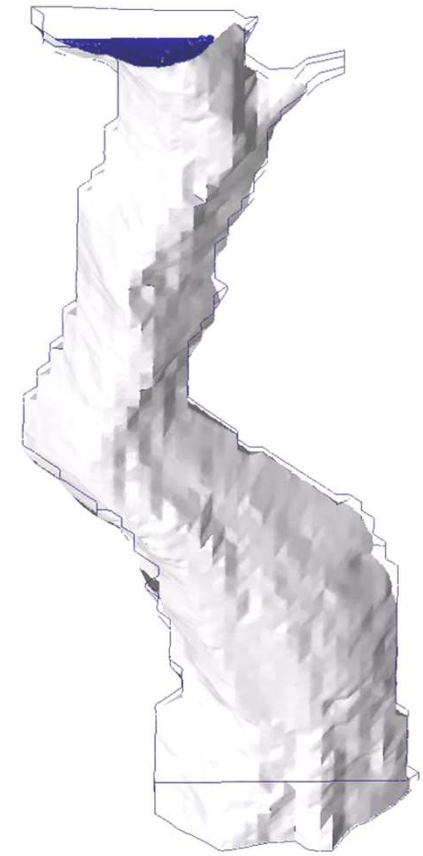
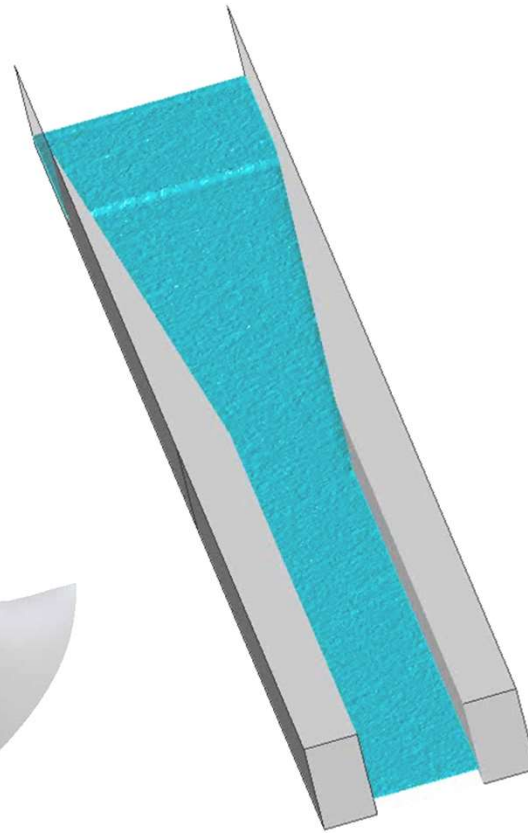
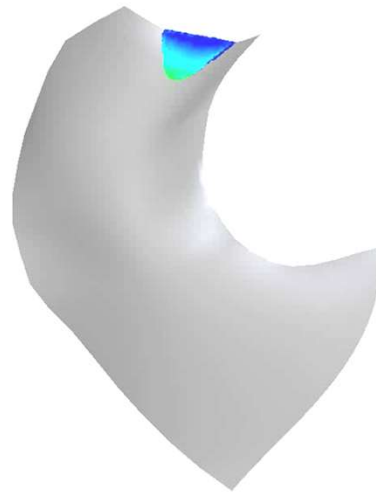
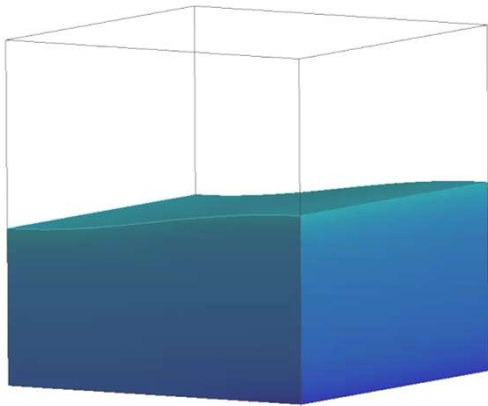
2-phase and free surface flow



2-phase and free surface flow

2-phase solver (air + whatever):

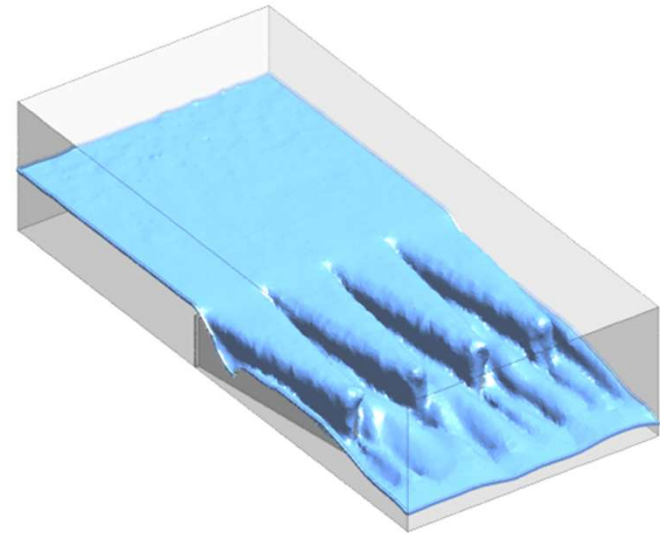
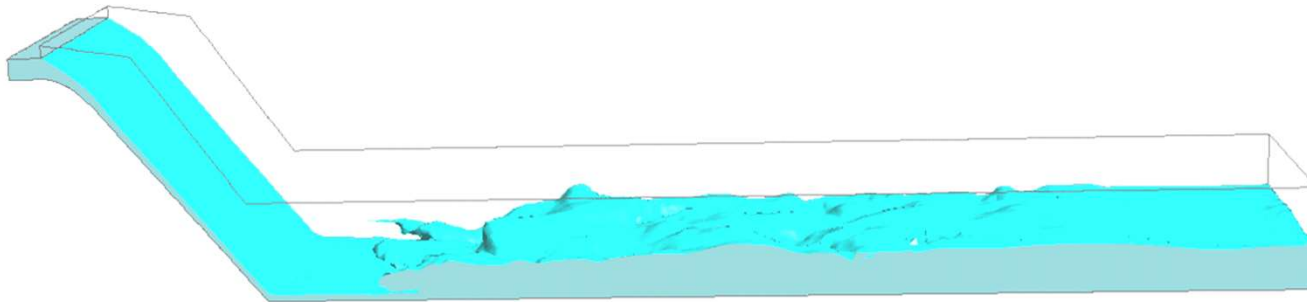
- Level set-based
- VMS stabilisation
- Pressure enrichment
- Mass and energy conservation



2-phase and free surface flow

Free-surface solver:

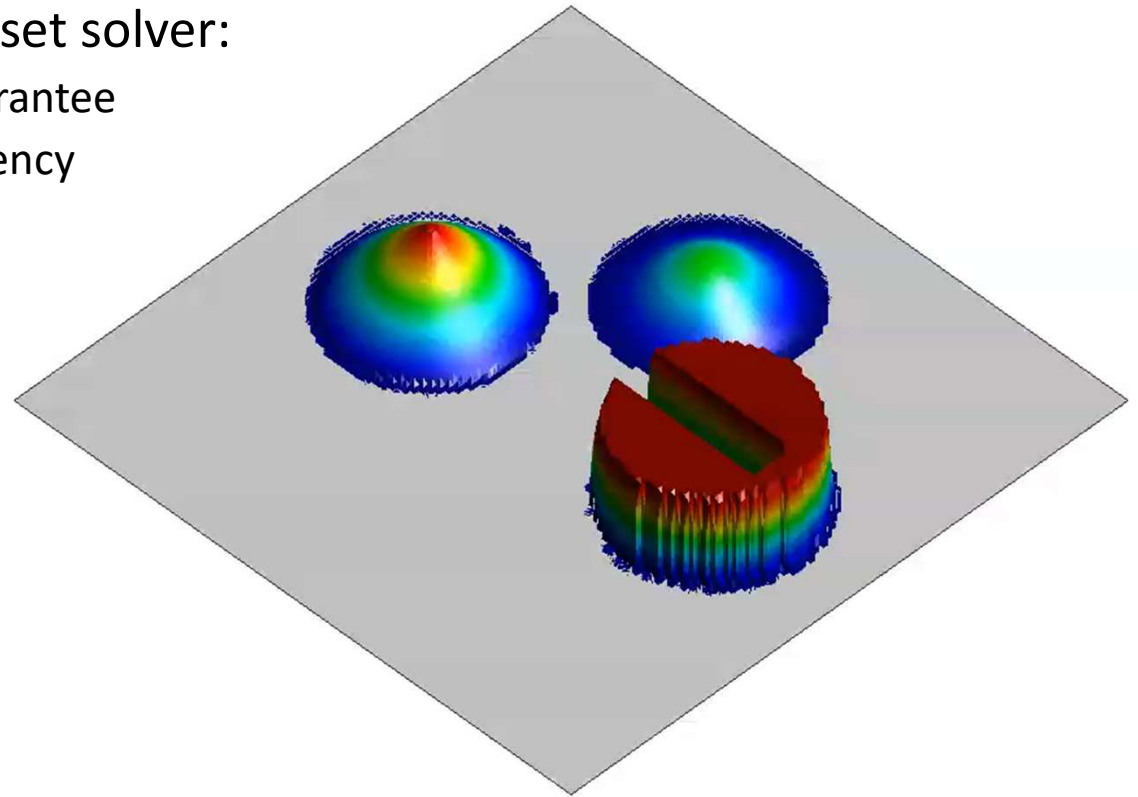
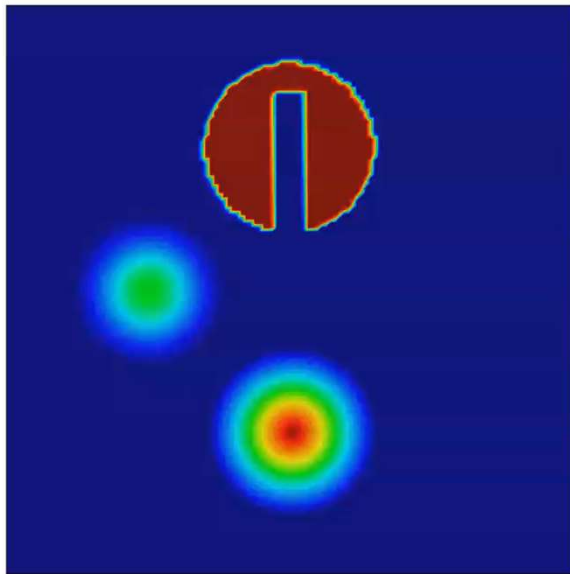
- Level set-based
- Porous media flow coupling
- Edge-based implementation



2-phase and free surface flow

Flux Corrected Transport (FCT) level set solver:

- Algebraic flux correction $\rightarrow$ DMP guarantee
- Edge-based implementation $\rightarrow$ efficiency

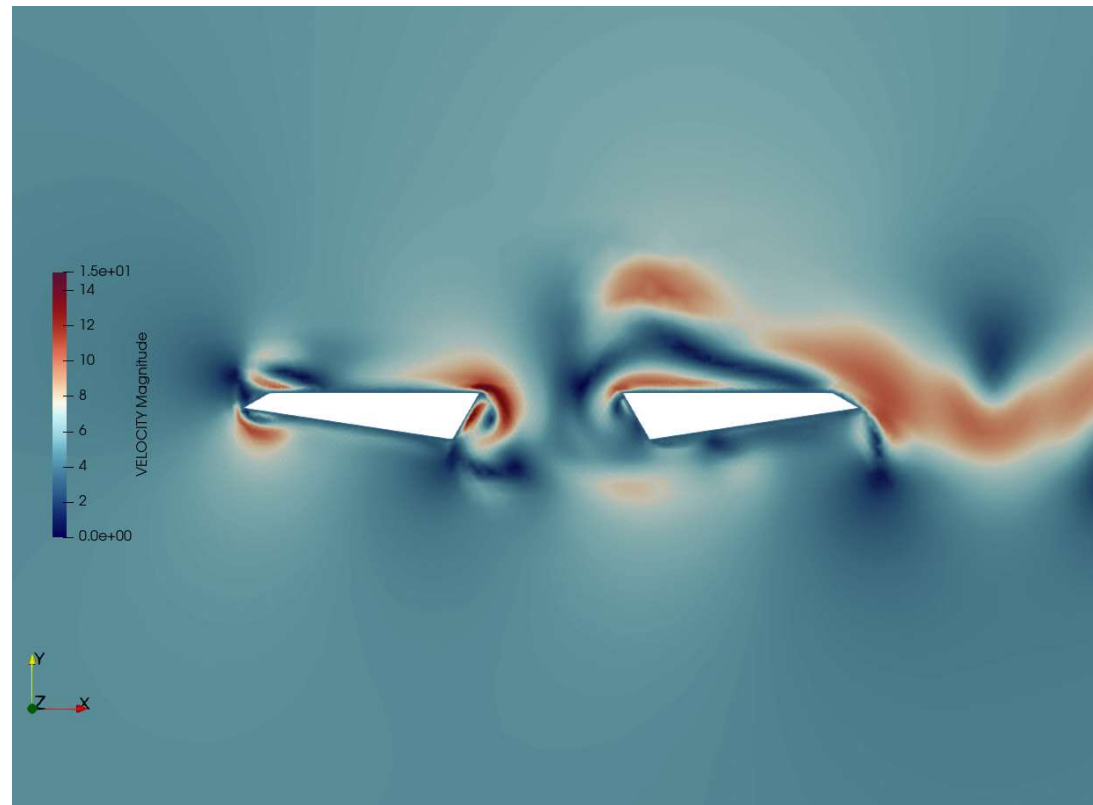


RANS and turbulence modelling



URANS and optimisation

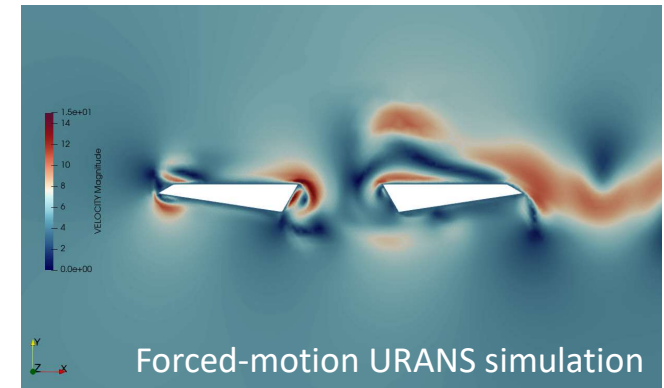
- URANS
- Turbulence models:
 - $k - \omega$
 - $k - \varepsilon$
 - $k - \omega - SST$
- Wall laws:
 - Logarithmic wall law
- Available sensitivities:
 - Frozen turbulence
 - Full sensitivities



URANS and optimisation

Flutter derivatives from forced-motion simulations:

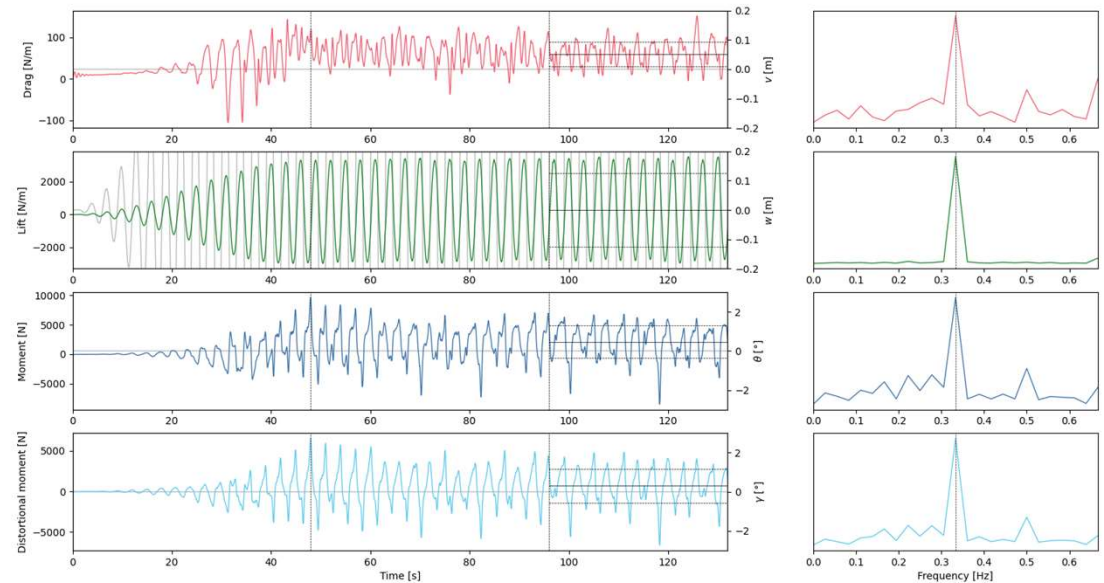
- **Cross-section aerodynamics**
- A sinusoidal motion imposition → **ALE**
- High number of runs (**efficiency**) → **URANS**



Cross-section resultant forces record



Flutter derivatives by curve fitting
to these time series



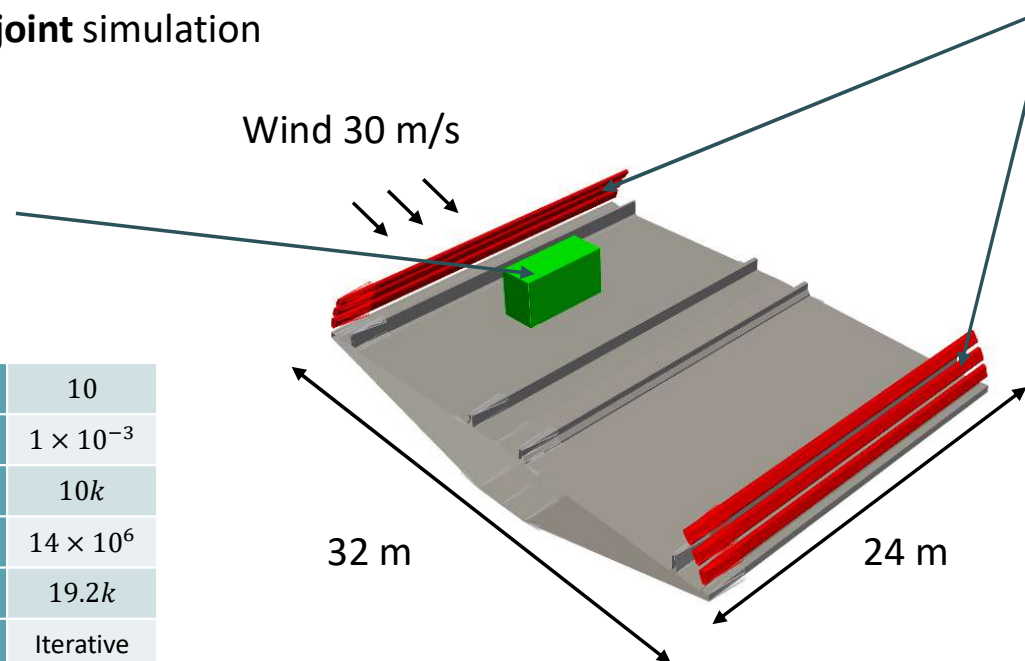
URANS and optimisation

Millau viaduct design optimisation

- **Transient primal** simulation
- **Transient adjoint** simulation

Truck	
Length (m)	6
Height (m)	3.28
Width (m)	2.5

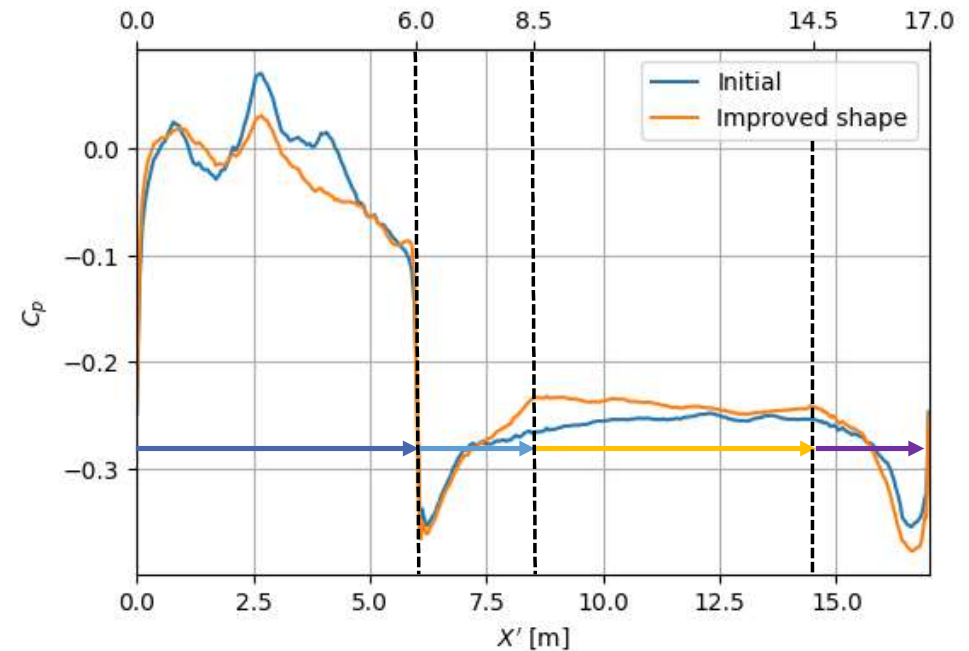
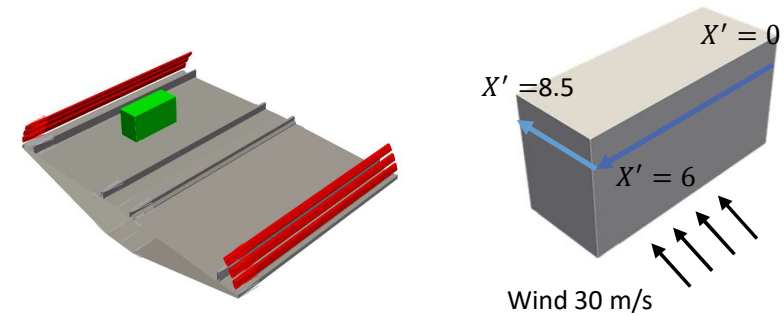
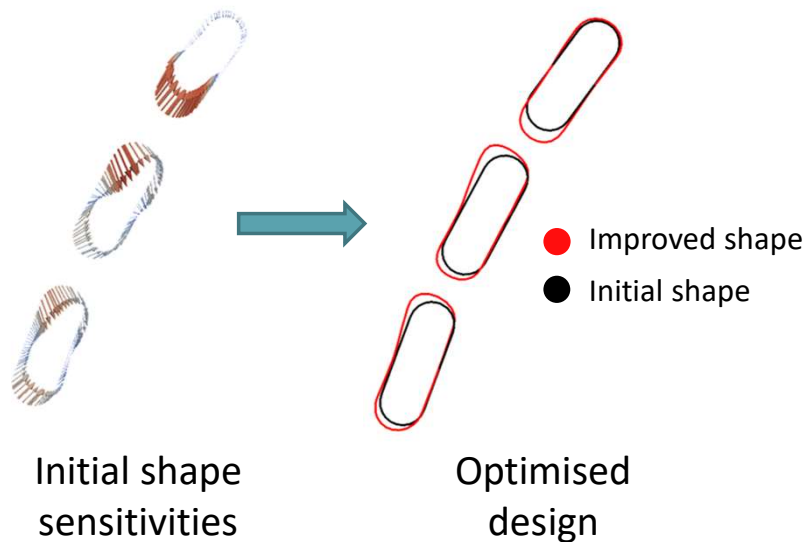
Total time (s)	10
Delta time (s)	1×10^{-3}
Time steps	10k
# of elements	14×10^6
Computation cost (h)	19.2k
Solver type	Iterative
Time avg. window (s)	[1, 10]



URANS and optimisation

Millau viaduct updated design

- **Objective:** minimize drag on truck
- **Design surface:** 398k design variables



Truck c_p distribution at 3m height



Other technologies

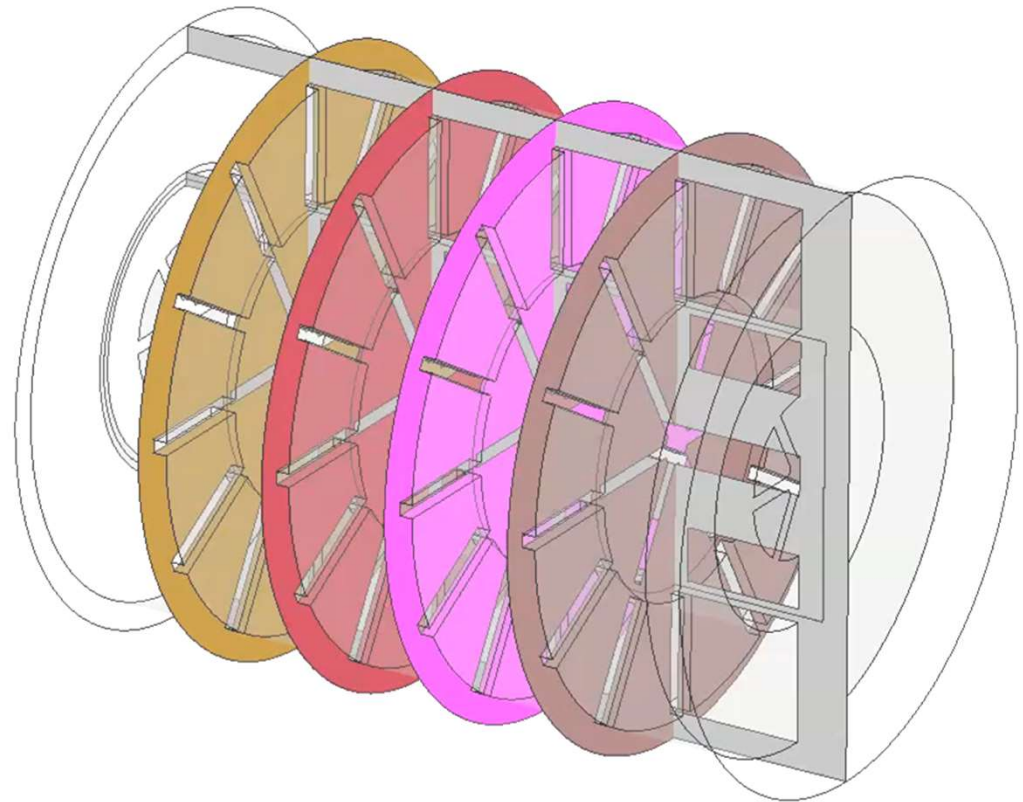
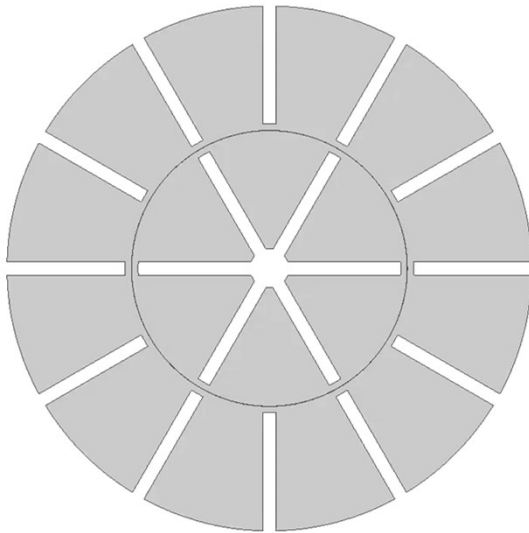
Sliding mesh, adaptive mesh refinement and geomodelling



Other technologies: master-slave constraints

Master-slave constraints:

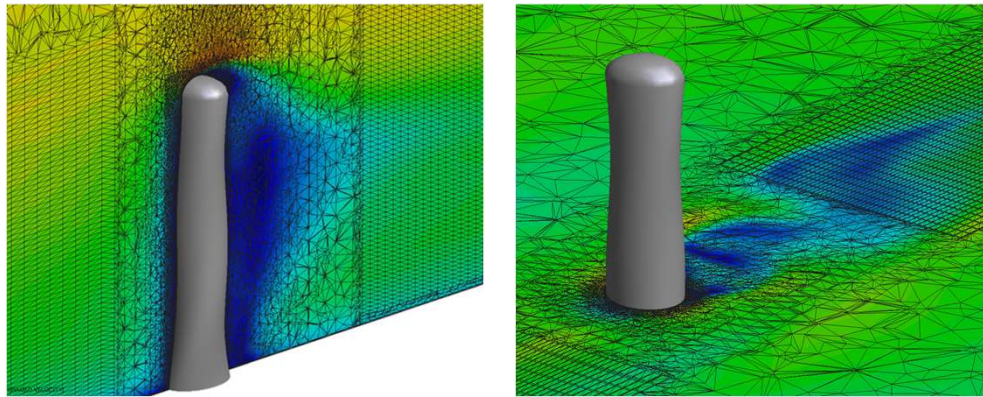
- Sliding mesh methods
- Periodicity
- ...



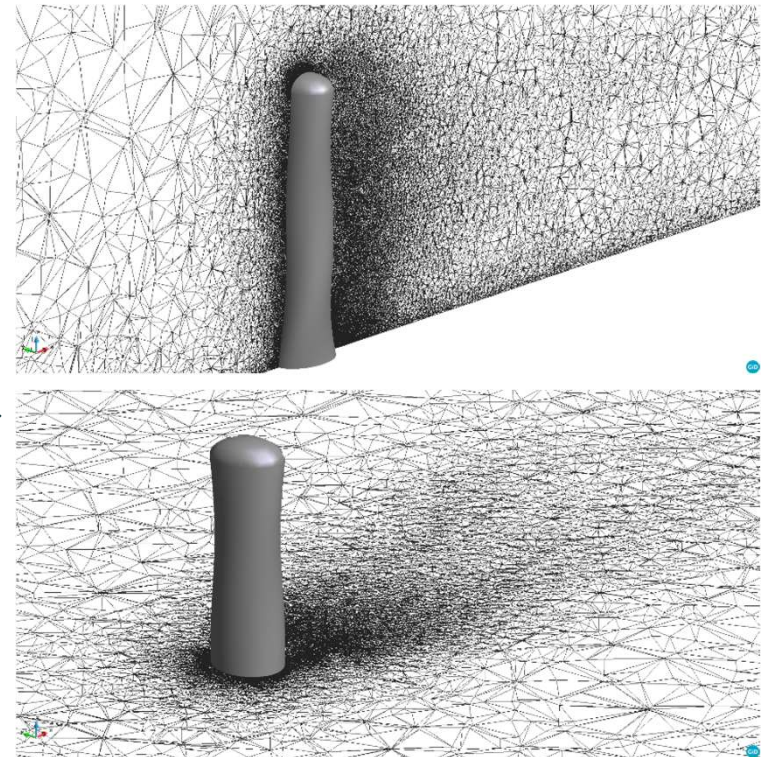
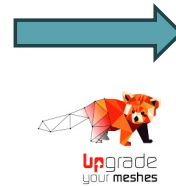
Other technologies: adaptive mesh refinement

Adaptive Mesh Refinement (AMR):

- Kratos quantities $\rightarrow$ metric calculation
- AMR $\rightarrow$ interface to MMG



Initial mesh (96 procs, 3M elements)



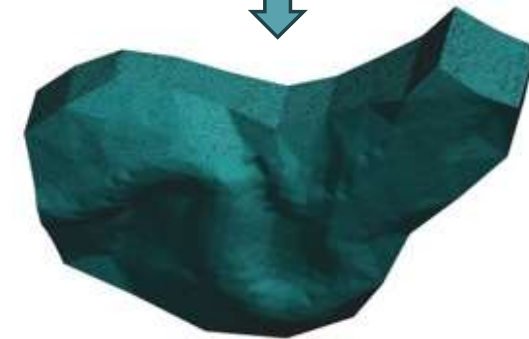
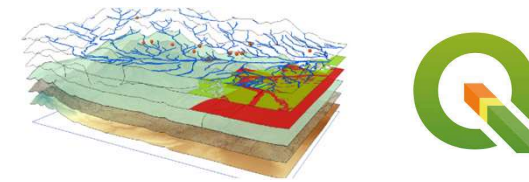
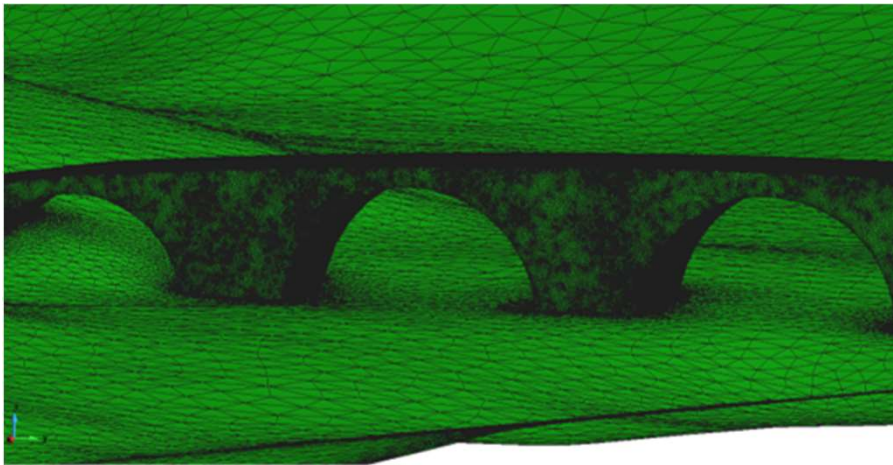
Adapted mesh (96 procs, 5M elements)



Other technologies: geomodelling

- **Geomodeling tool**

- Automatic 3D mesh from GIS data
- Automatic object insertion (structures)
- Multipurpose (wind, flooding, ...)



Coupling

Fluid-Structure Interaction and thermal



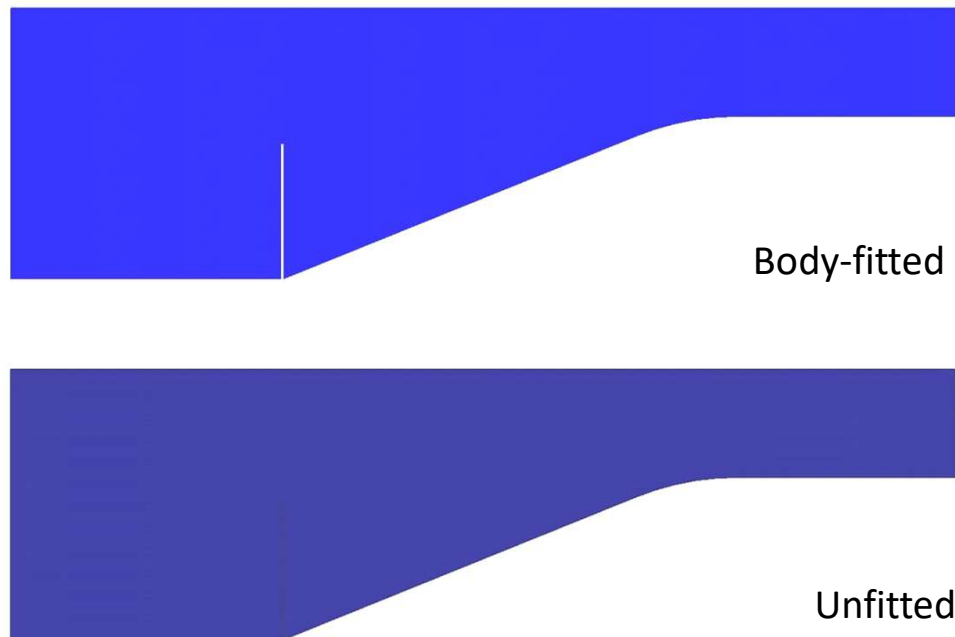
Coupling: Fluid-Structure Interaction

- Partitioned coupling strategy
 - **Dirichlet – Neuman coupling**
 - **Gauss-Seidel iteration** → Strongly coupled problems
 - OpenMP / MPI support
- Convergence accelerators
 - **Relaxation**
 - **Quasi-Newton methods**
 - **MVQN** and **IBQN-MVQN**
 - **Randomized MVQN** and **Randomized IBQN-MVQN**
- Capability to couple with other codes (a.k.a. CoSimulation)

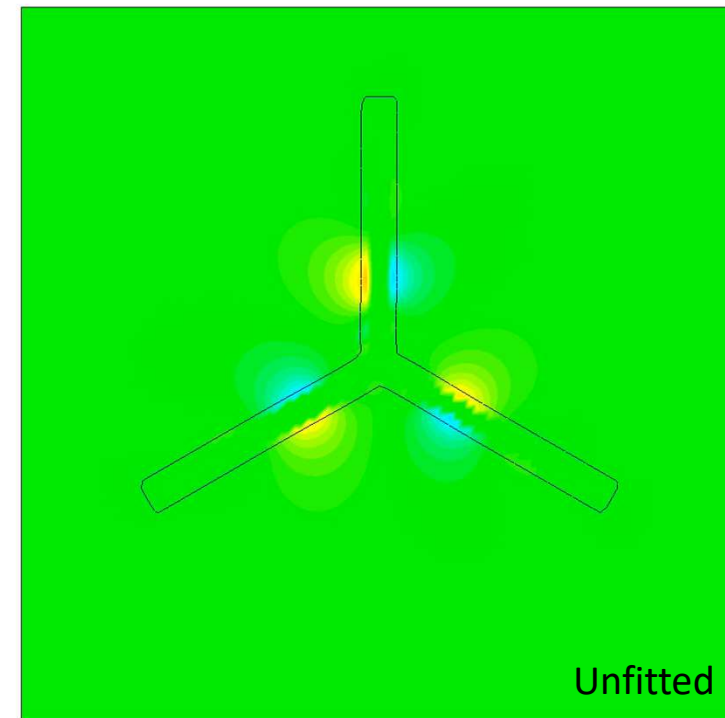


Coupling: Fluid-Structure Interaction

2D Mok benchmark



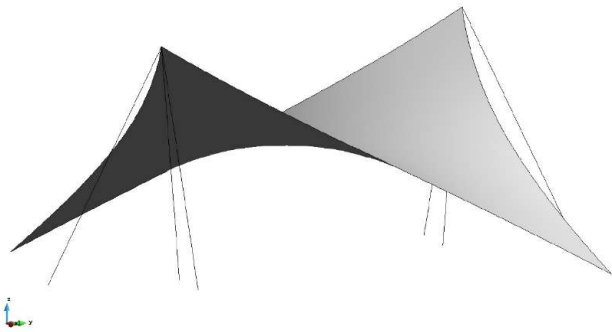
2D Mixer w/ flexible blades



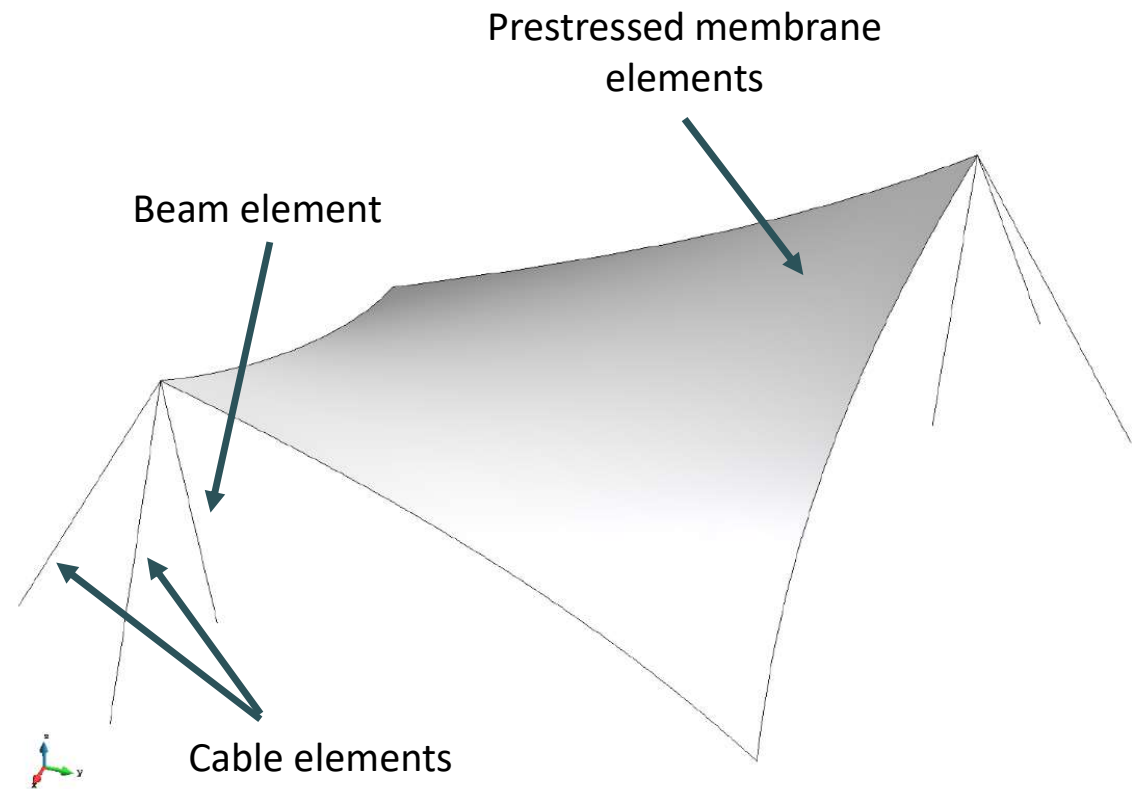
Coupling: Fluid-Structure Interaction

3D 4-point tent

- 100x40x20m wind channel
- Hyperbolic paraboloid membrane
 - Plan size: 8x8
 - Elevation: 4
- Storm wind → Inlet: 50m/s



6/11/2024 – 8/11/2024

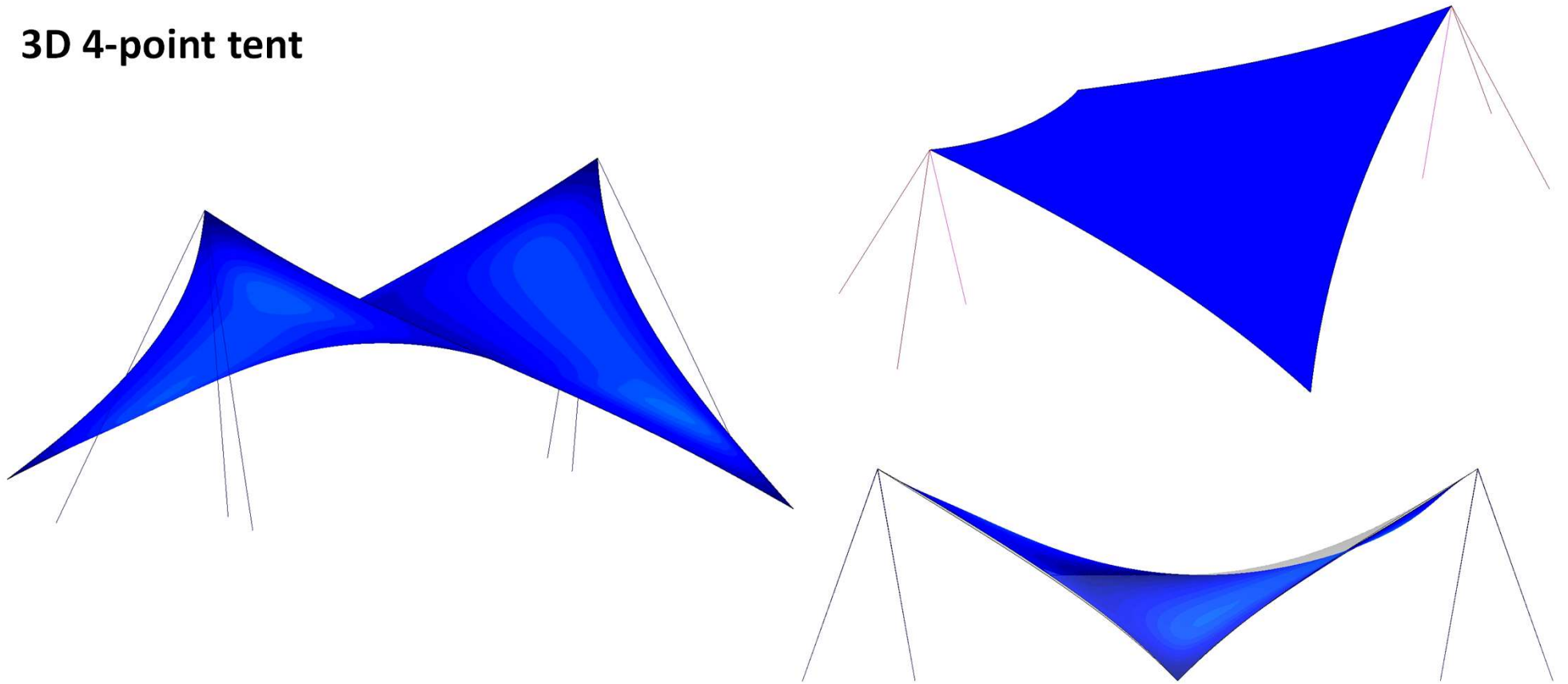


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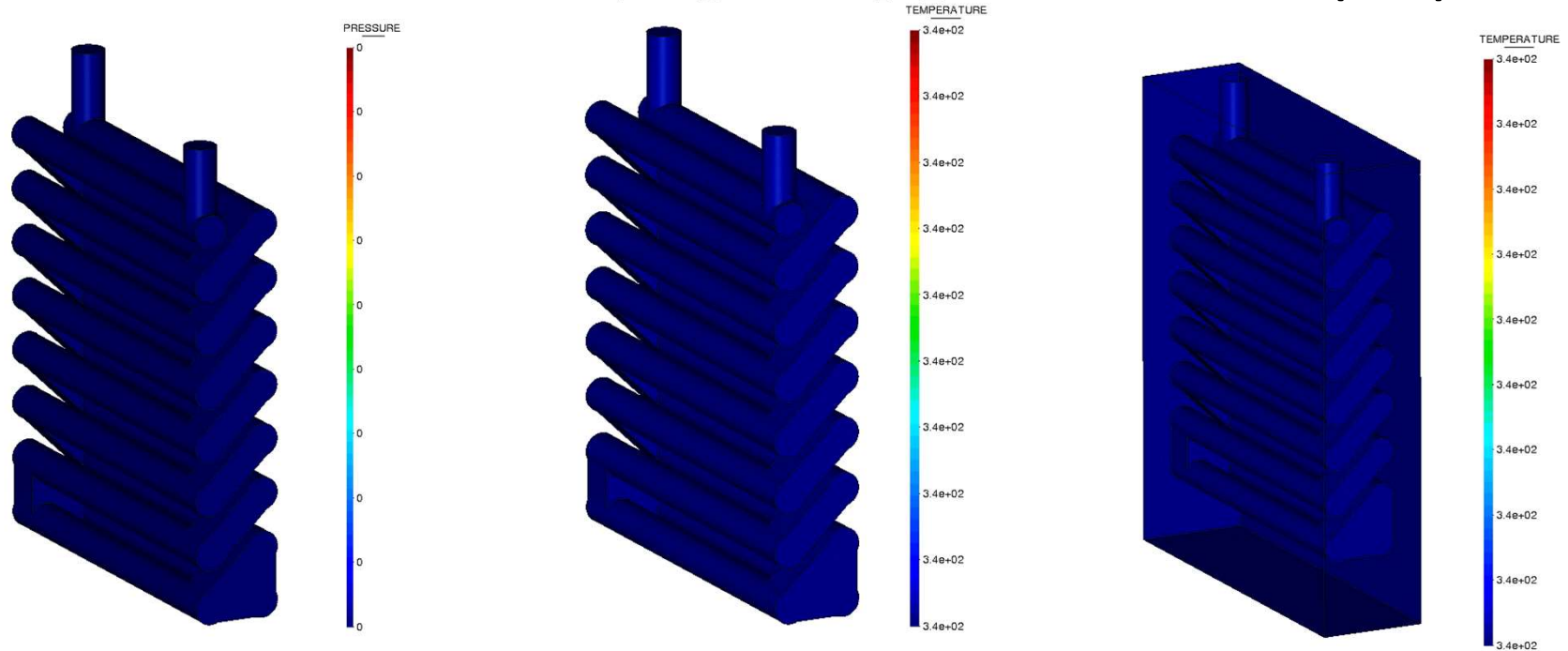
Coupling: Fluid-Structure Interaction

3D 4-point tent



Coupling: thermal

- Volumetric **fluid-fluid** coupling → **Buoyancy approach**
- Partitioned **fluid – structure** coupling → **Conjugate Heat Transfer (CHT)**



Ongoing research & developments

- Marco A. Zúñiga (@Marco1410) 
 - ROM of potential transonic flows → Shape optimization
 - Multi-fidelity simulation in aerodynamics
- Uxue Chasco (@uxuech) 
 - Splitting algorithms for mass and energy conservation in 2-phase flows
 - Hydraulic and torrent flows simulation
- Nicolò Antonelli (@NickNick9) 
 - Unfitted mesh methods in Isogeometric Analysis
 - Isogeometric Analysis in CFD applications
- Riccardo Rossi, Carlos Roig and Rubén Zorrilla (@RiccardoRossi, @roigcarlo and @rubenzorrilla) 
 - Targeting extremely large scale simulations → Fast structured GPU parallel solver





 <https://github.com/KratosMultiphysics>

 www.youtube.com/@kratosmultiphysics3578

